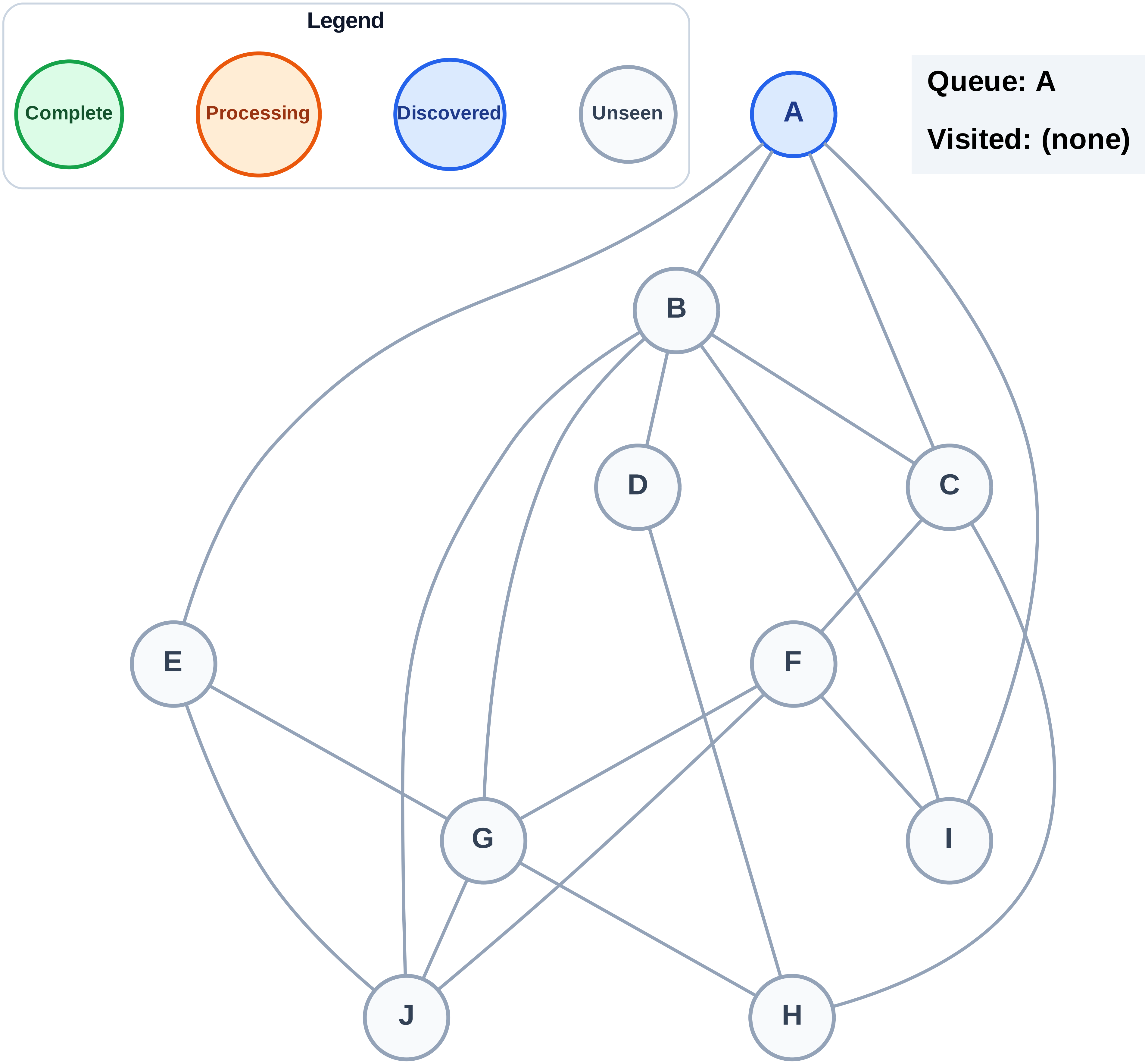
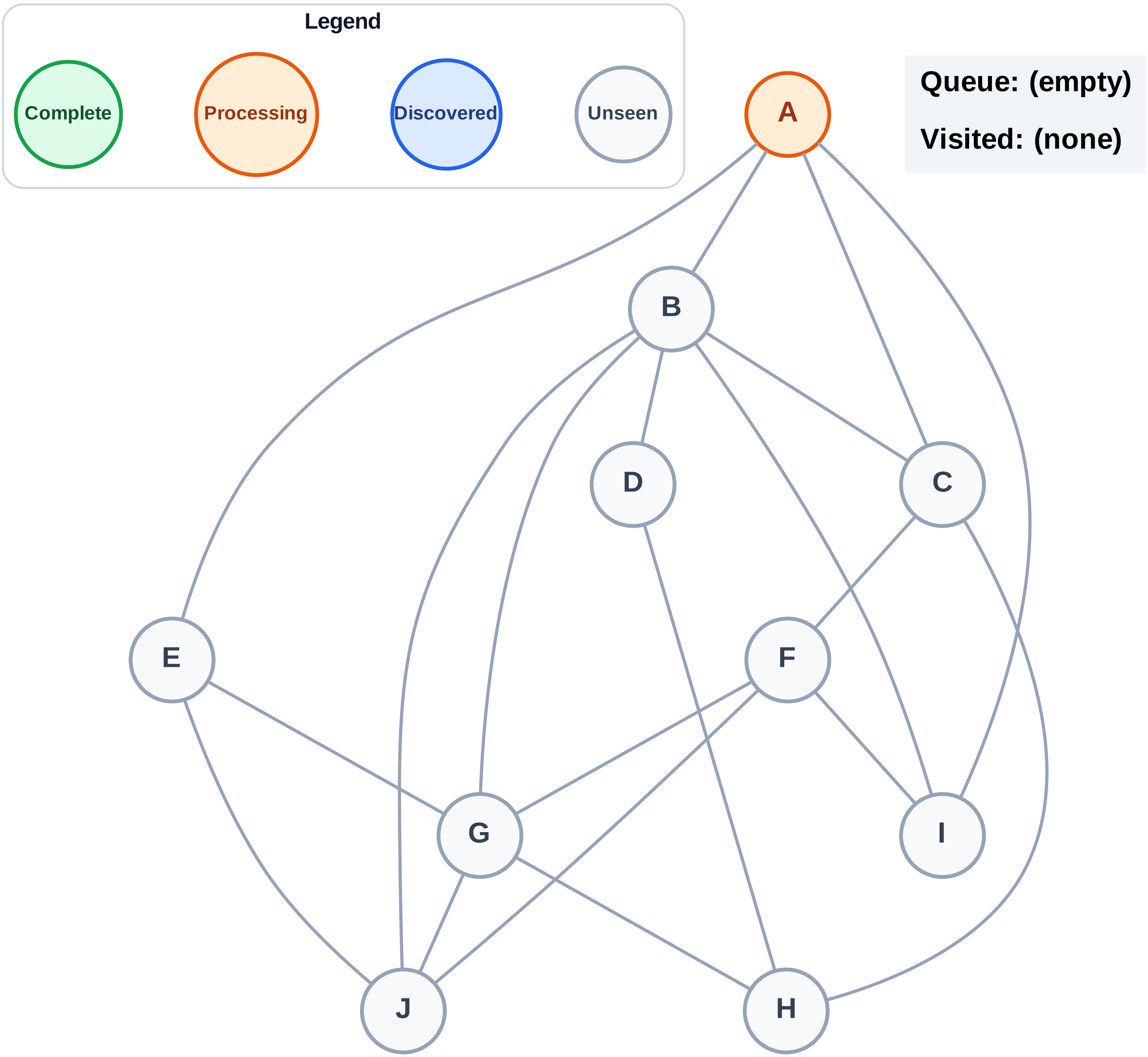
BFS Traversal

Step 1: Start at A



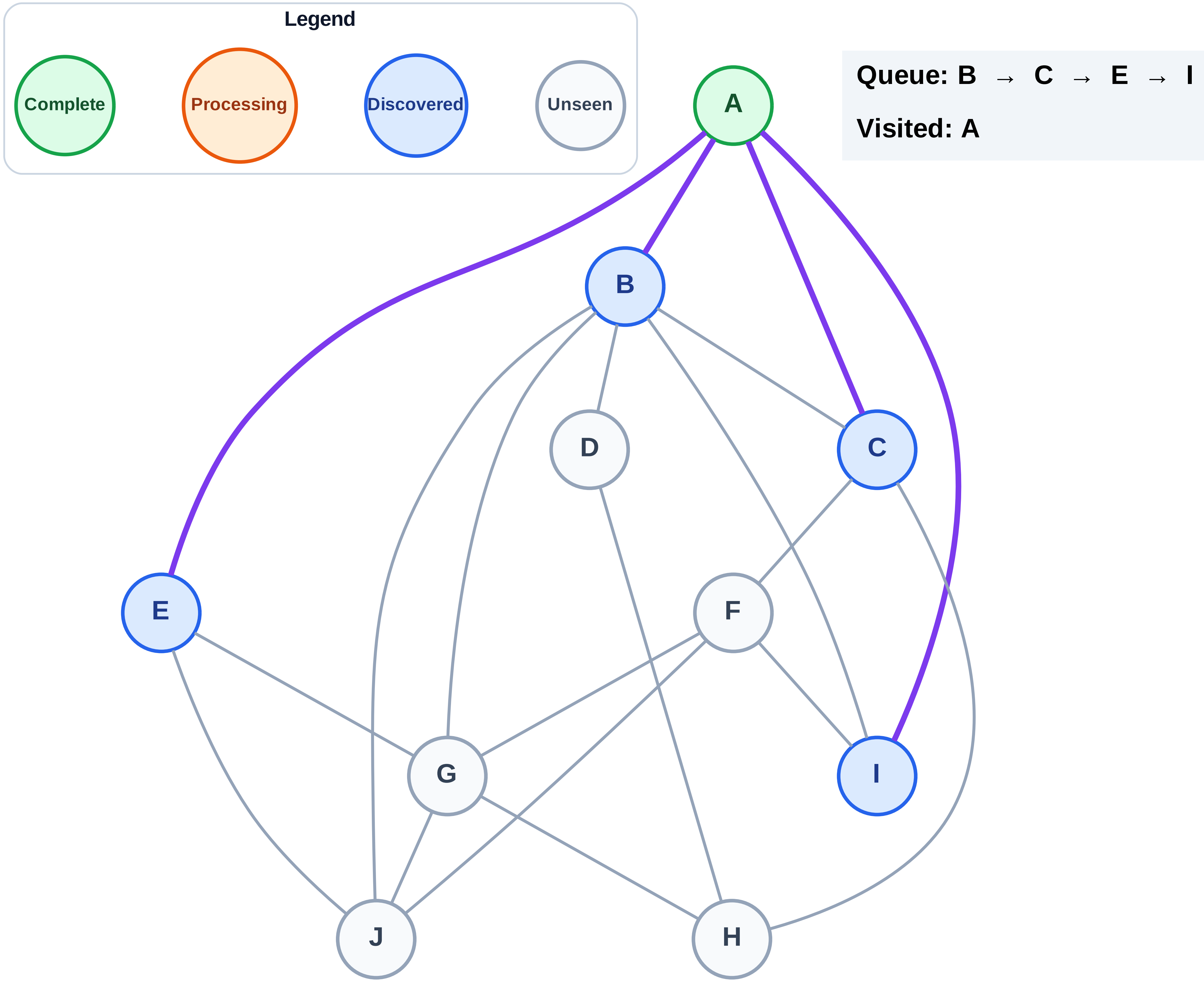
BFS Traversal

Step 2: Process node A



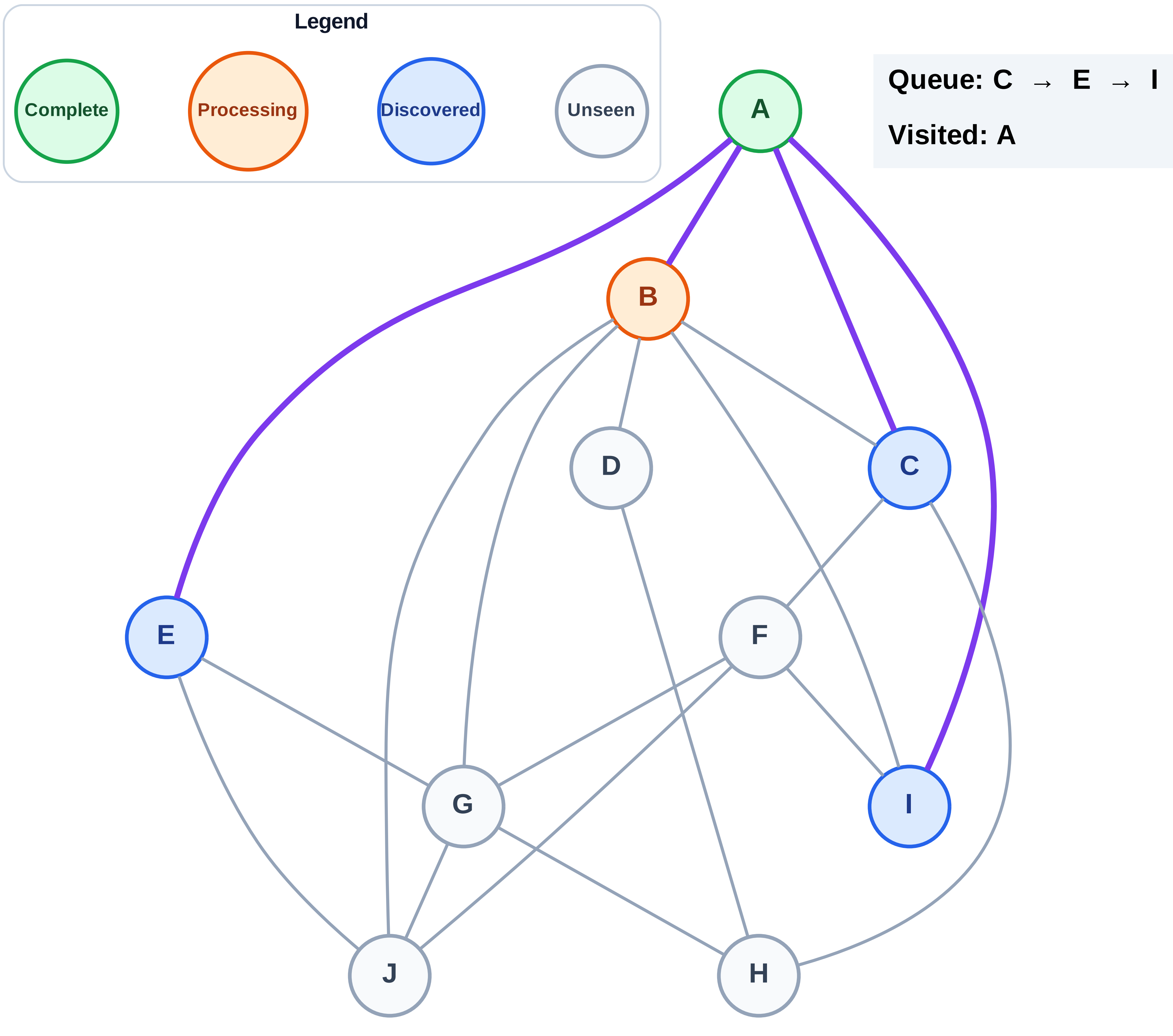
BFS Traversal

Step 3: Discover B, C, E, I from A



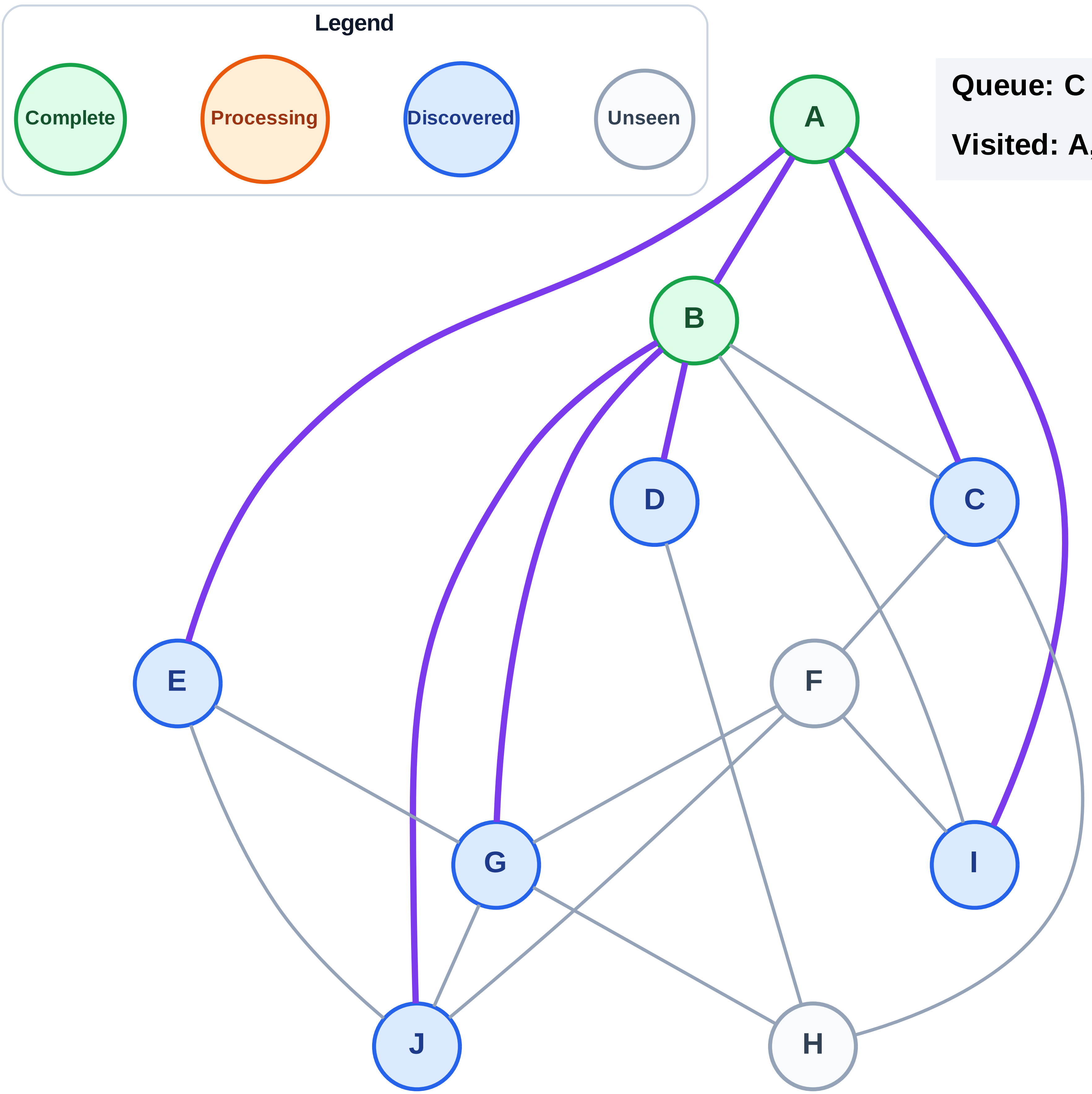
BFS Traversal

Step 4: Process node B



BFS Traversal

Step 5: Discover D, G, J from B

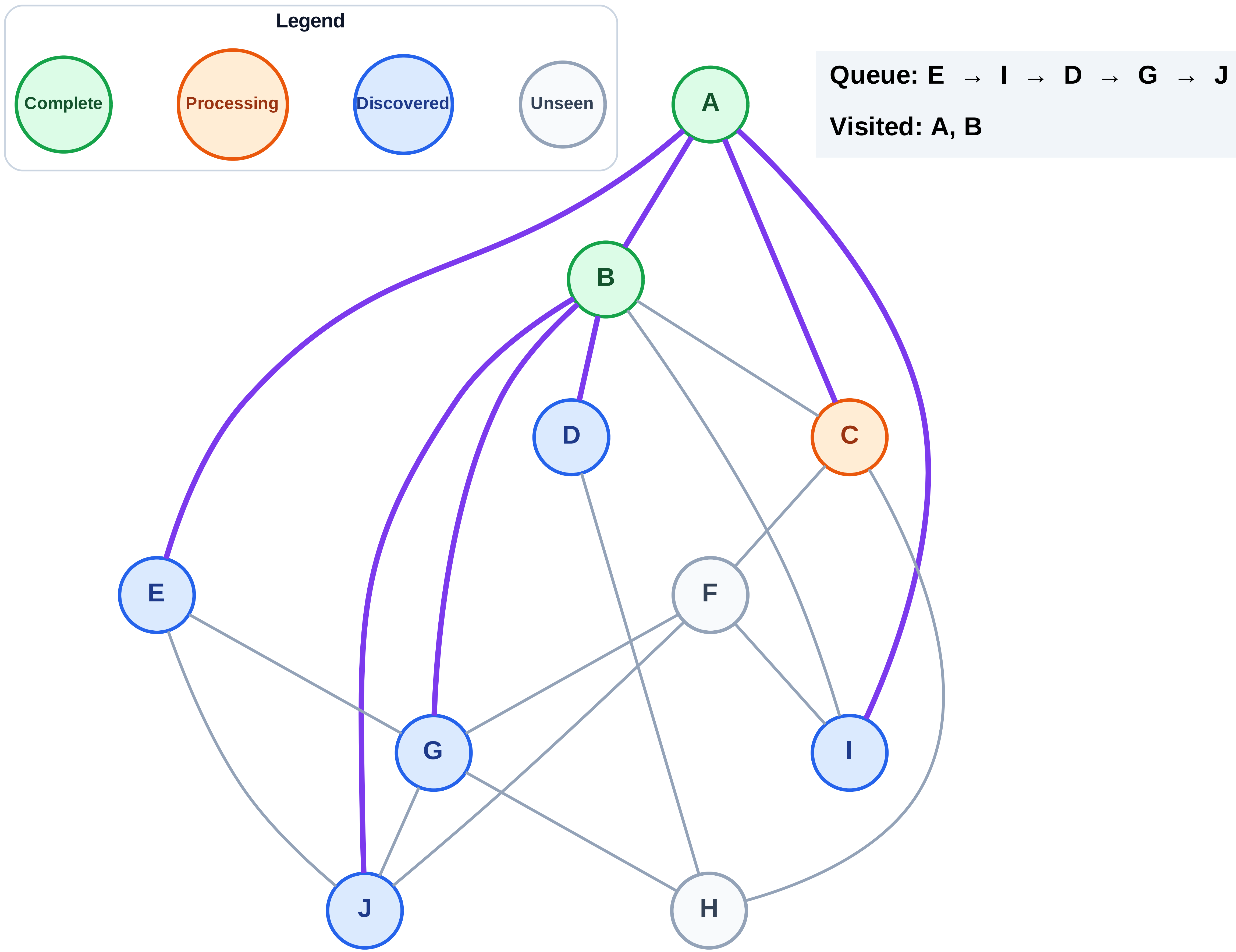


Queue: C → E → I → D → G → J

Visited: A, B

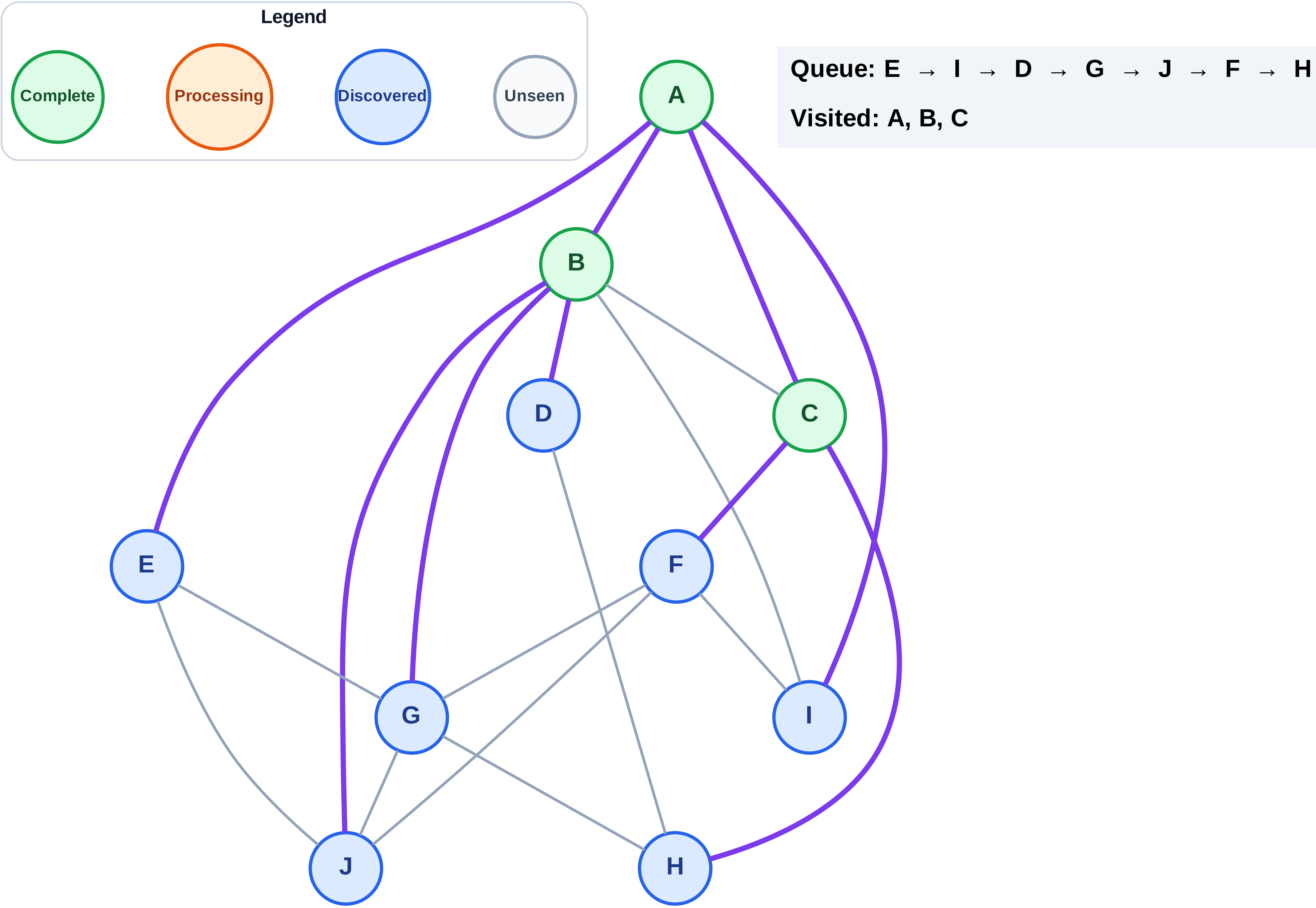
BFS Traversal

Step 6: Process node C



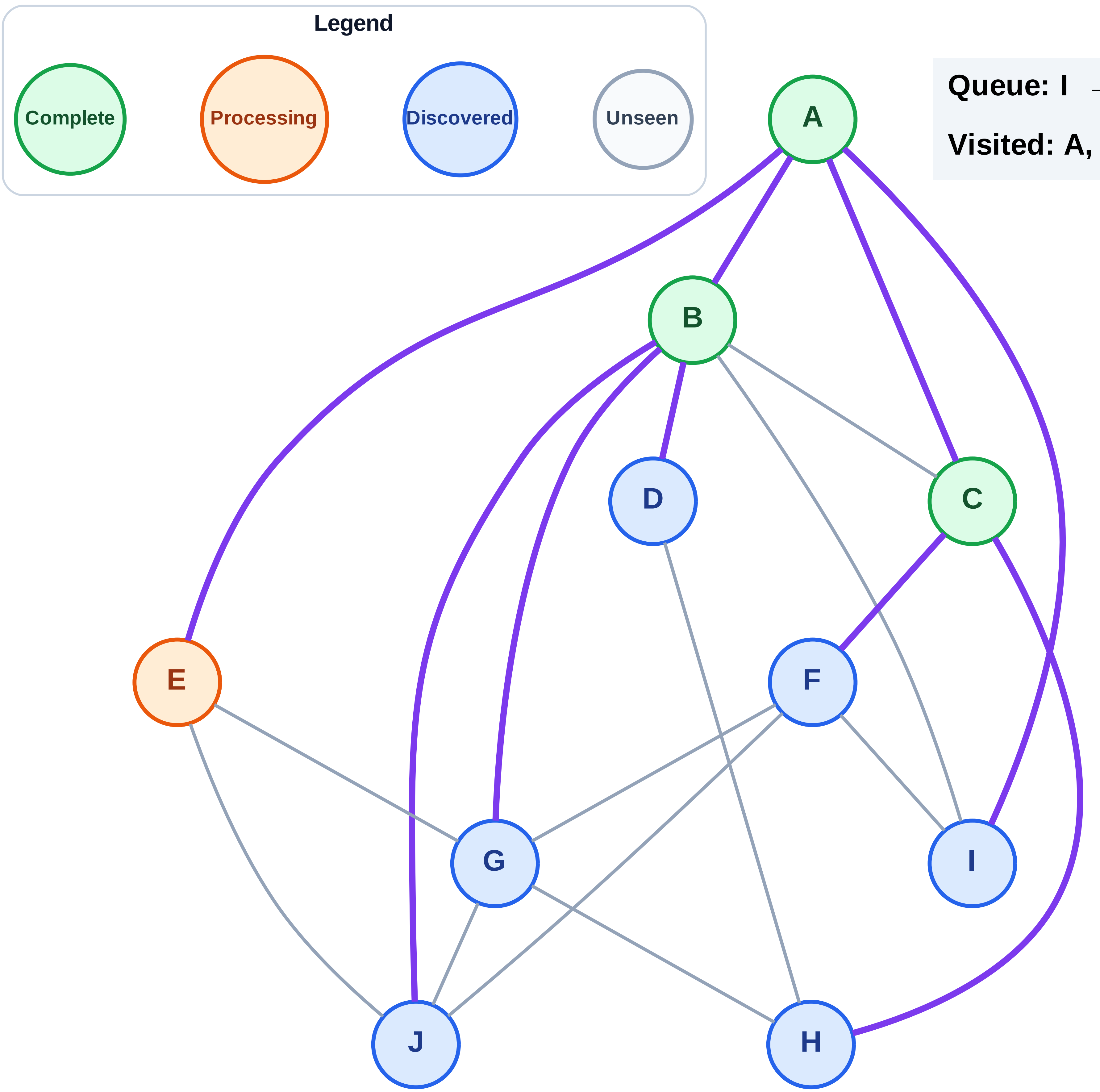
BFS Traversal

Step 7: Discover F, H from C



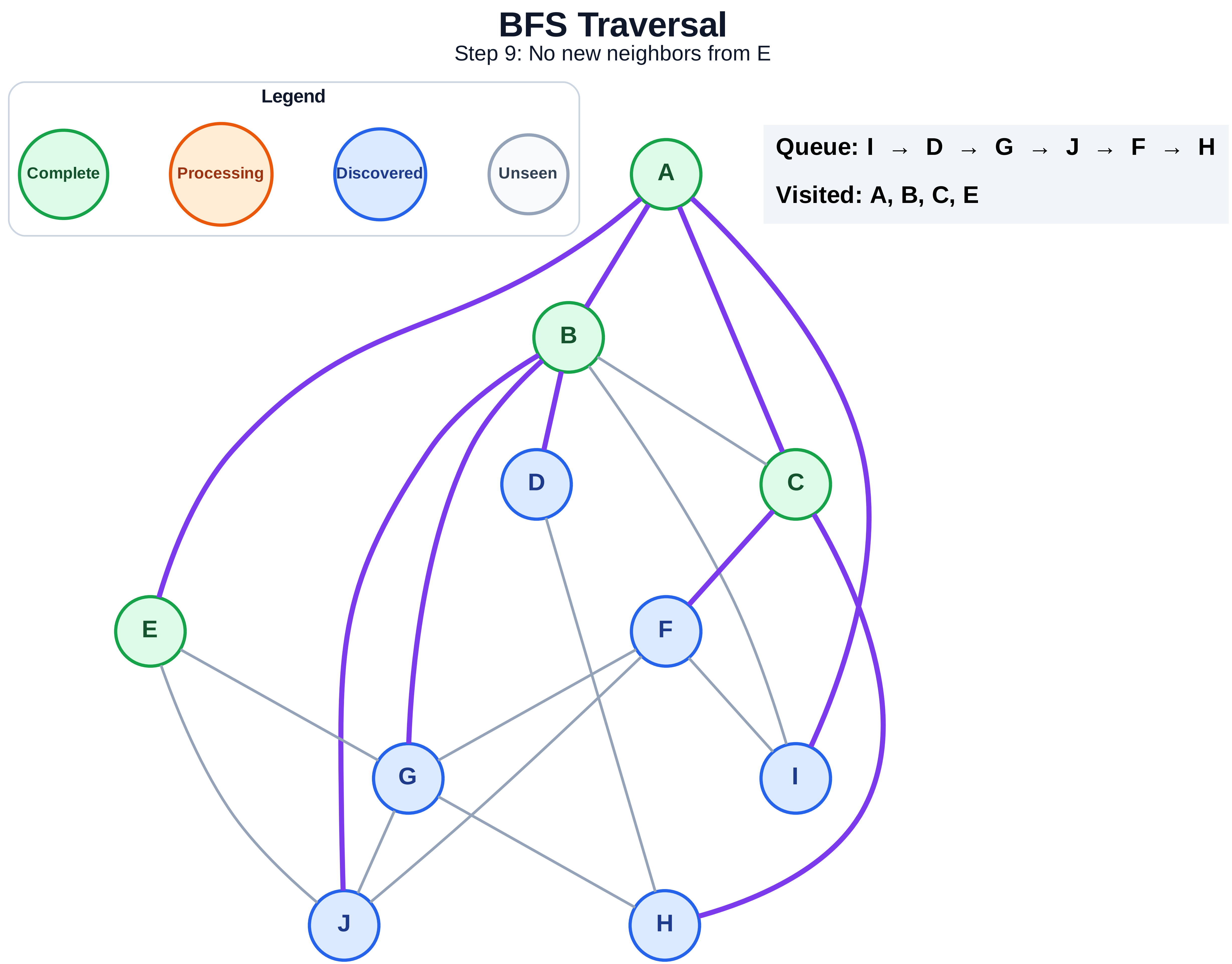
BFS Traversal

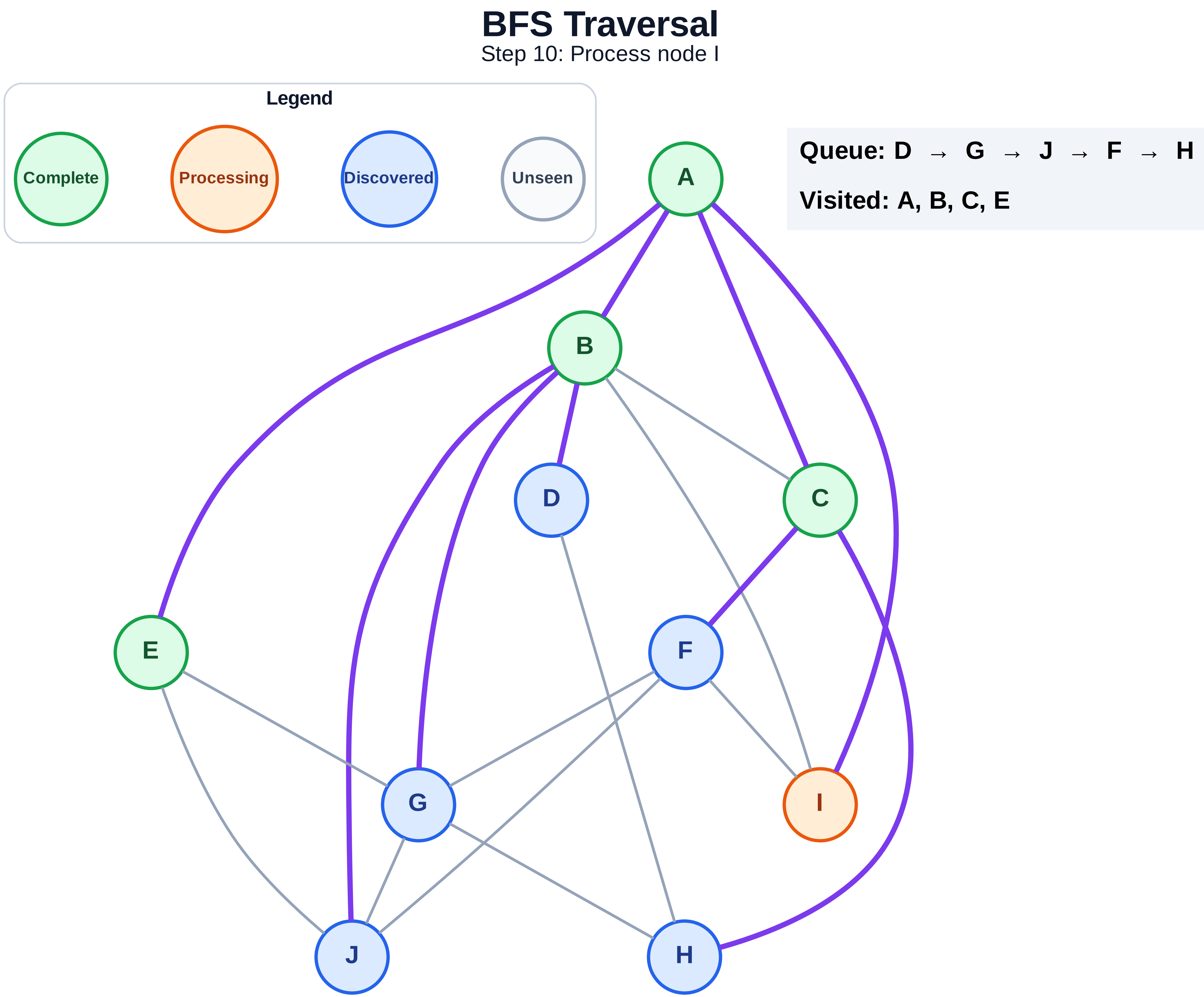
Step 8: Process node E



Queue: I → D → G → J → F → H

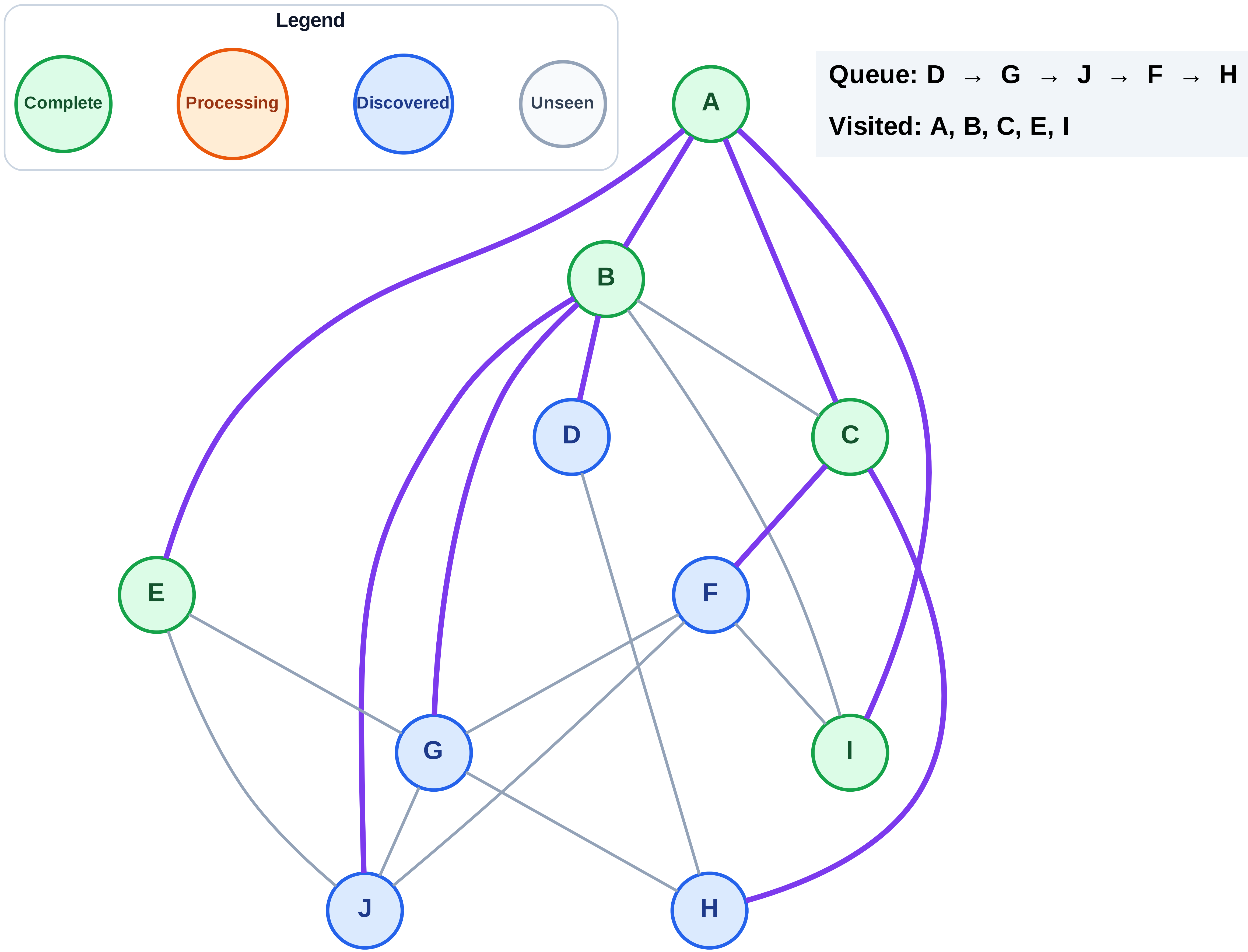
Visited: A, B, C


BFS Traversal
Step 9: No new neighbors from E
Legend
Complete
Processing
Discovered
Unseen
Queue: I → D → G → J → F → H
Visited: A, B, C, E
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 10: Process node I
Legend
Complete
Processing
Discovered
Unseen
Queue: D → G → J → F → H
Visited: A, B, C, E
A
B
C
D
E
F
G
H
I
J

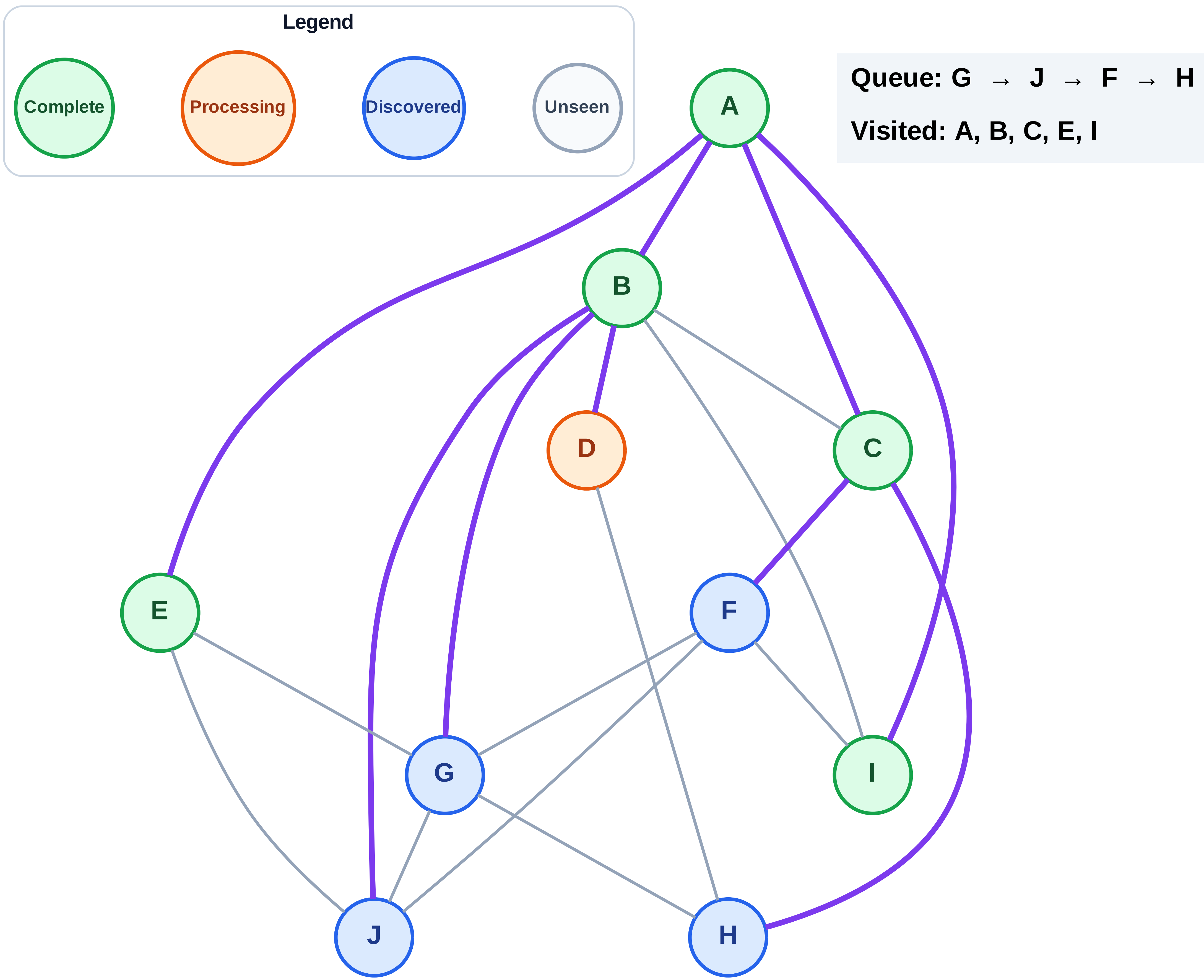
BFS Traversal

Step 11: No new neighbors from I



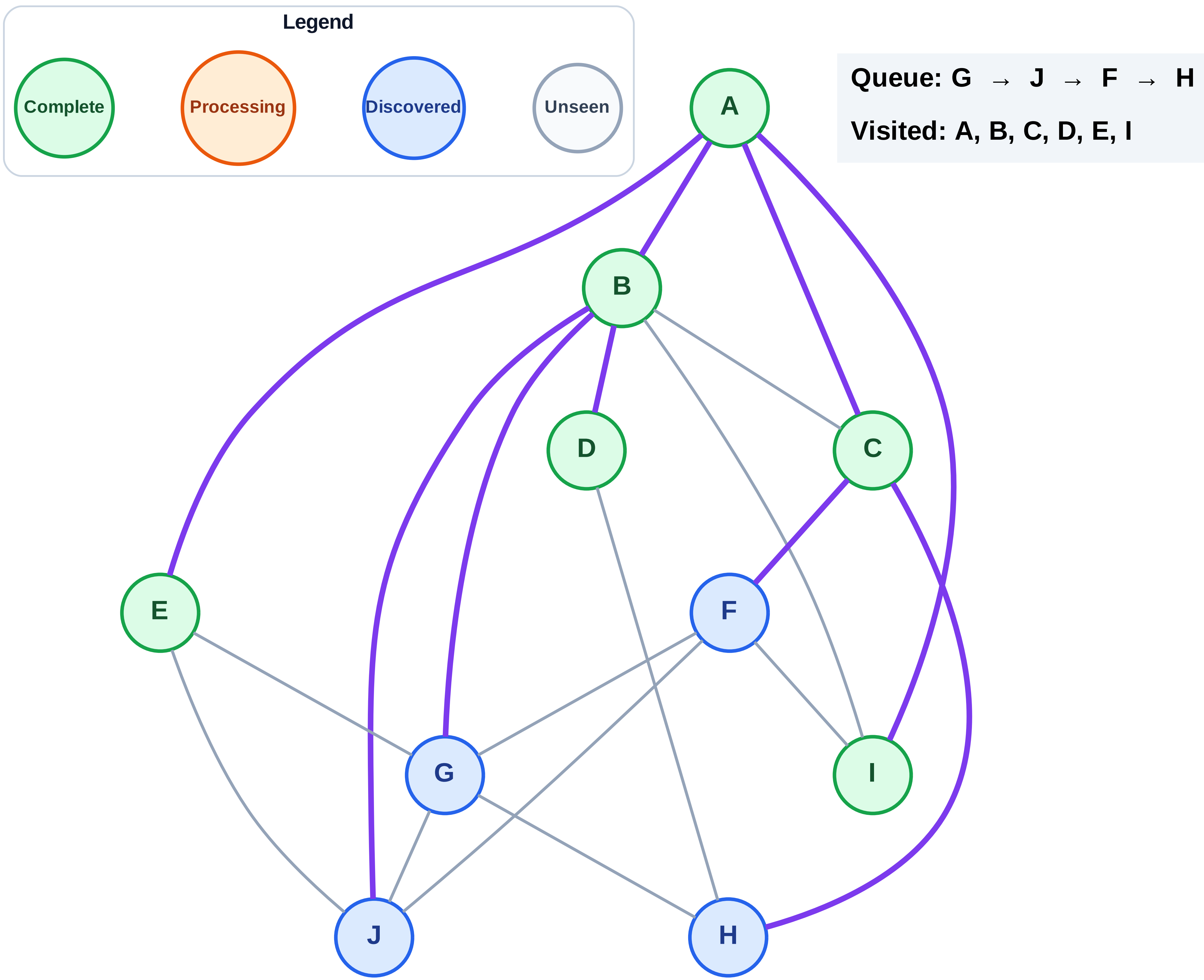
BFS Traversal

Step 12: Process node D



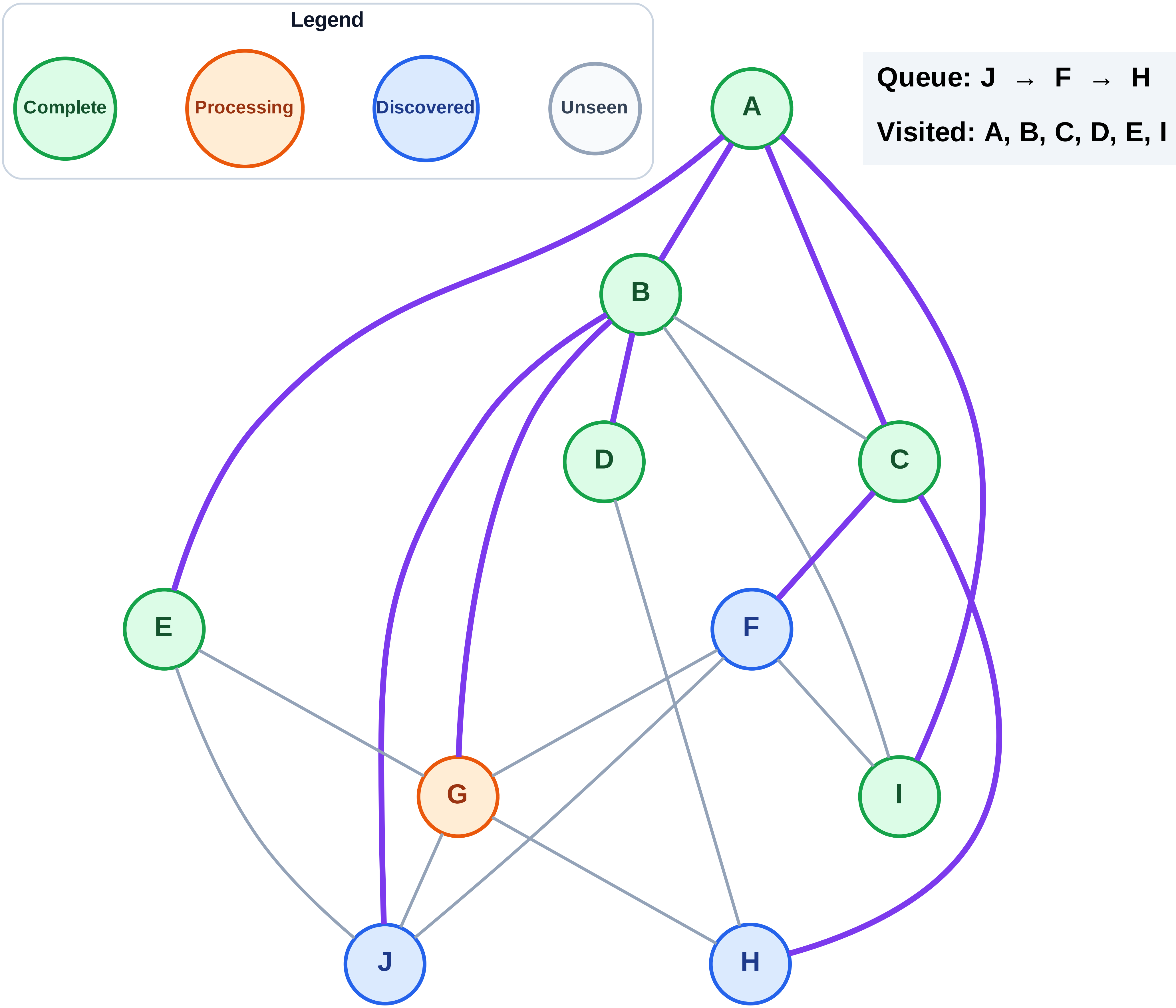
BFS Traversal

Step 13: No new neighbors from D



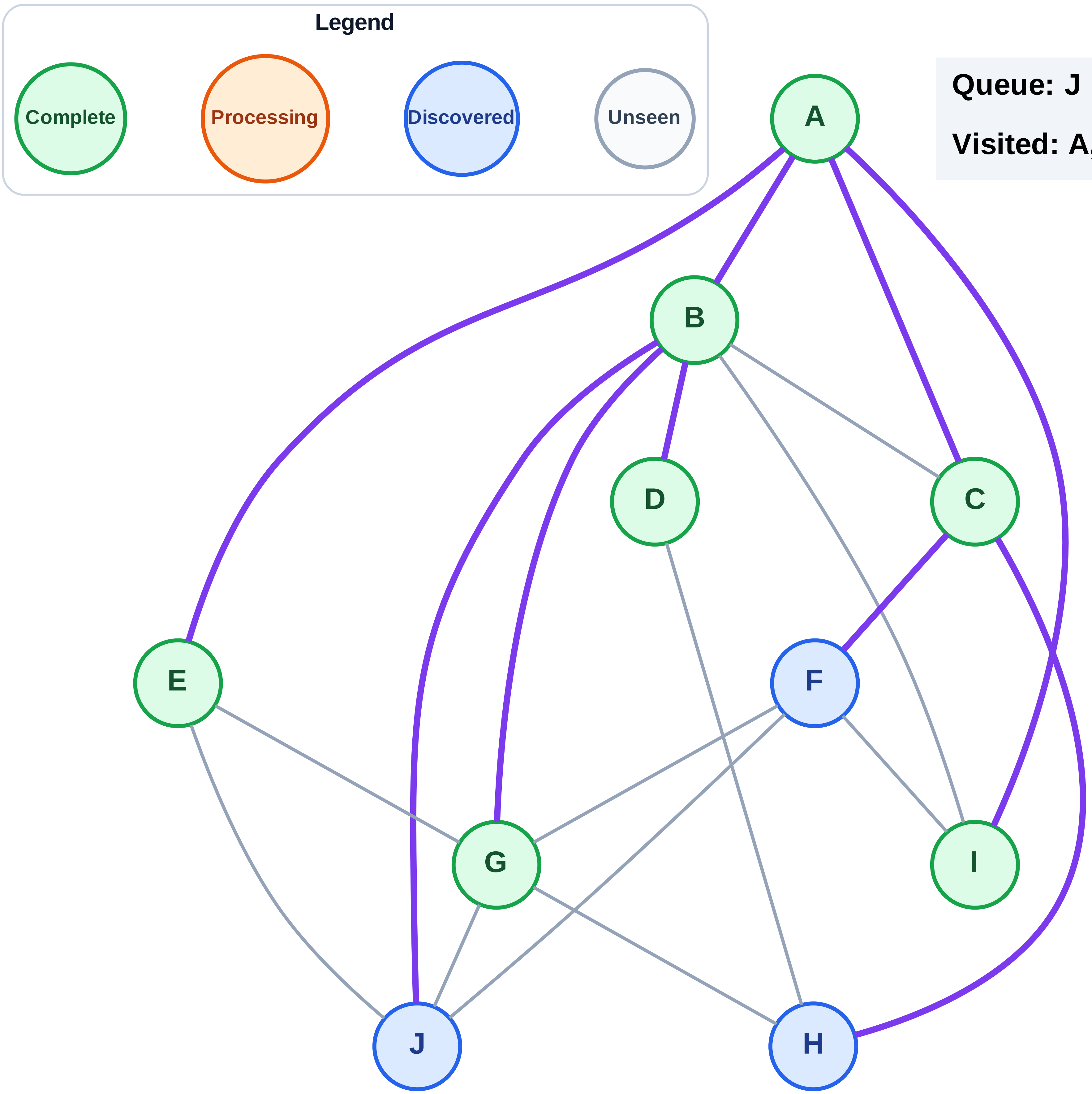
BFS Traversal

Step 14: Process node G



BFS Traversal

Step 15: No new neighbors from G

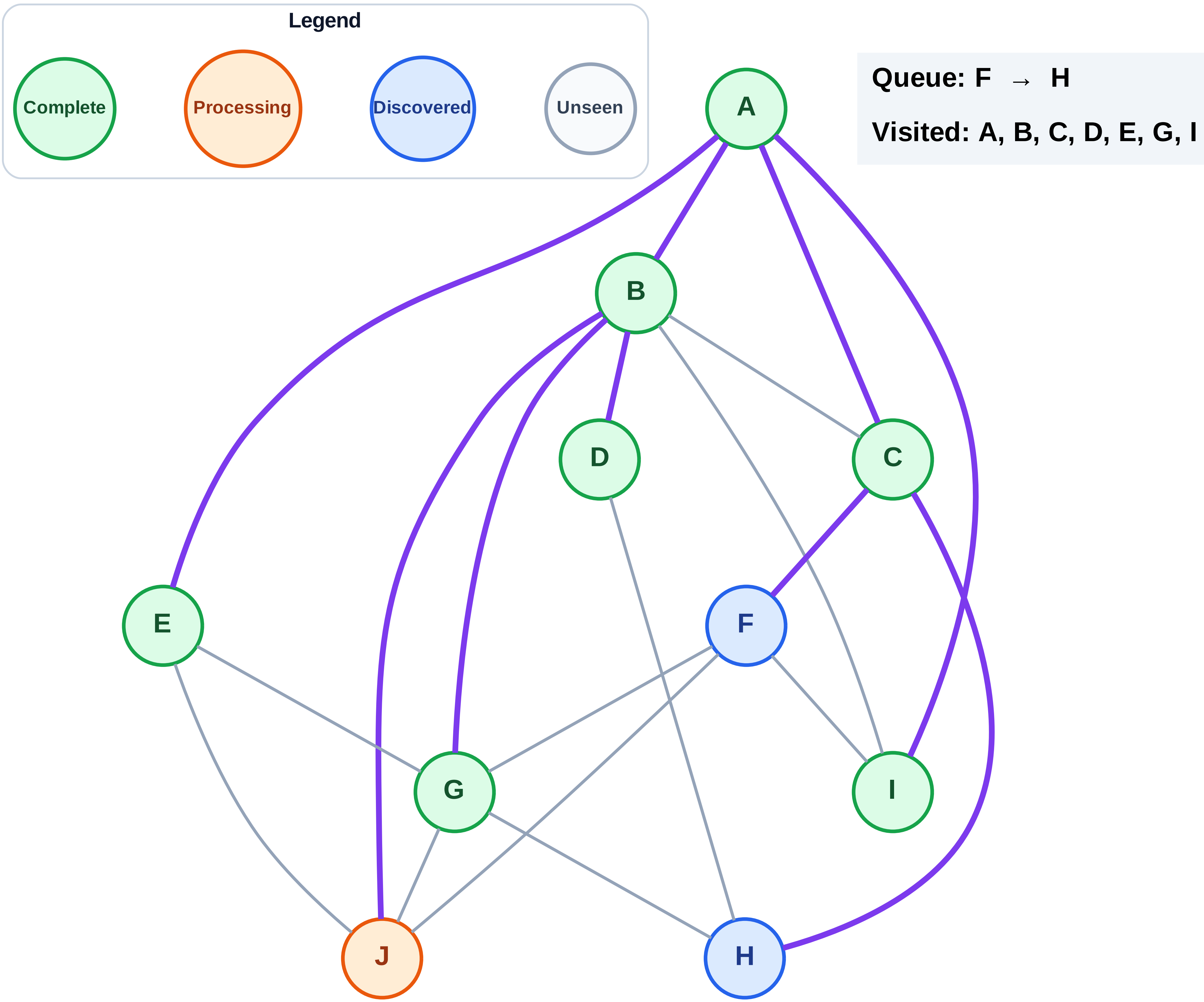


Queue: J → F → H

Visited: A, B, C, D, E, G, I

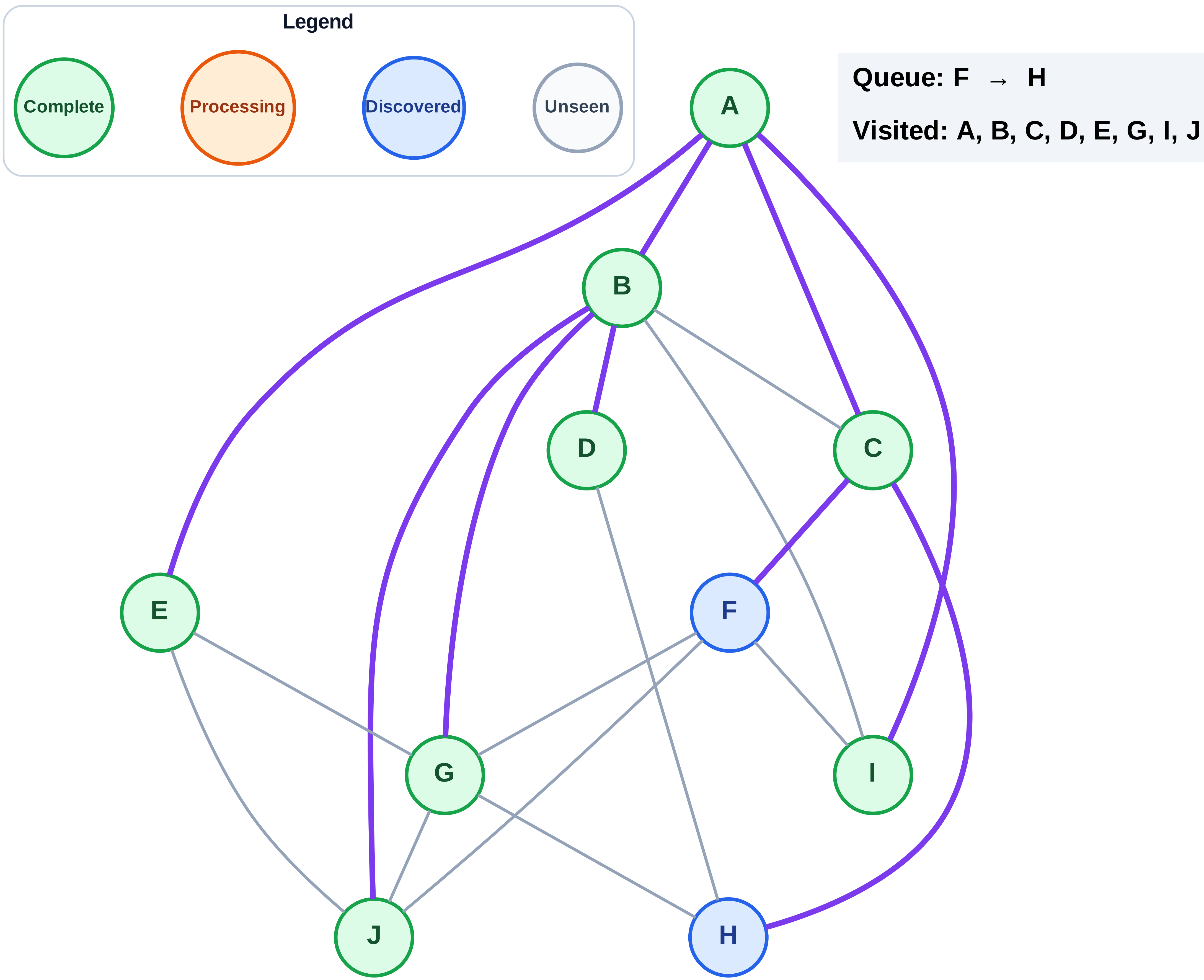
BFS Traversal

Step 16: Process node J



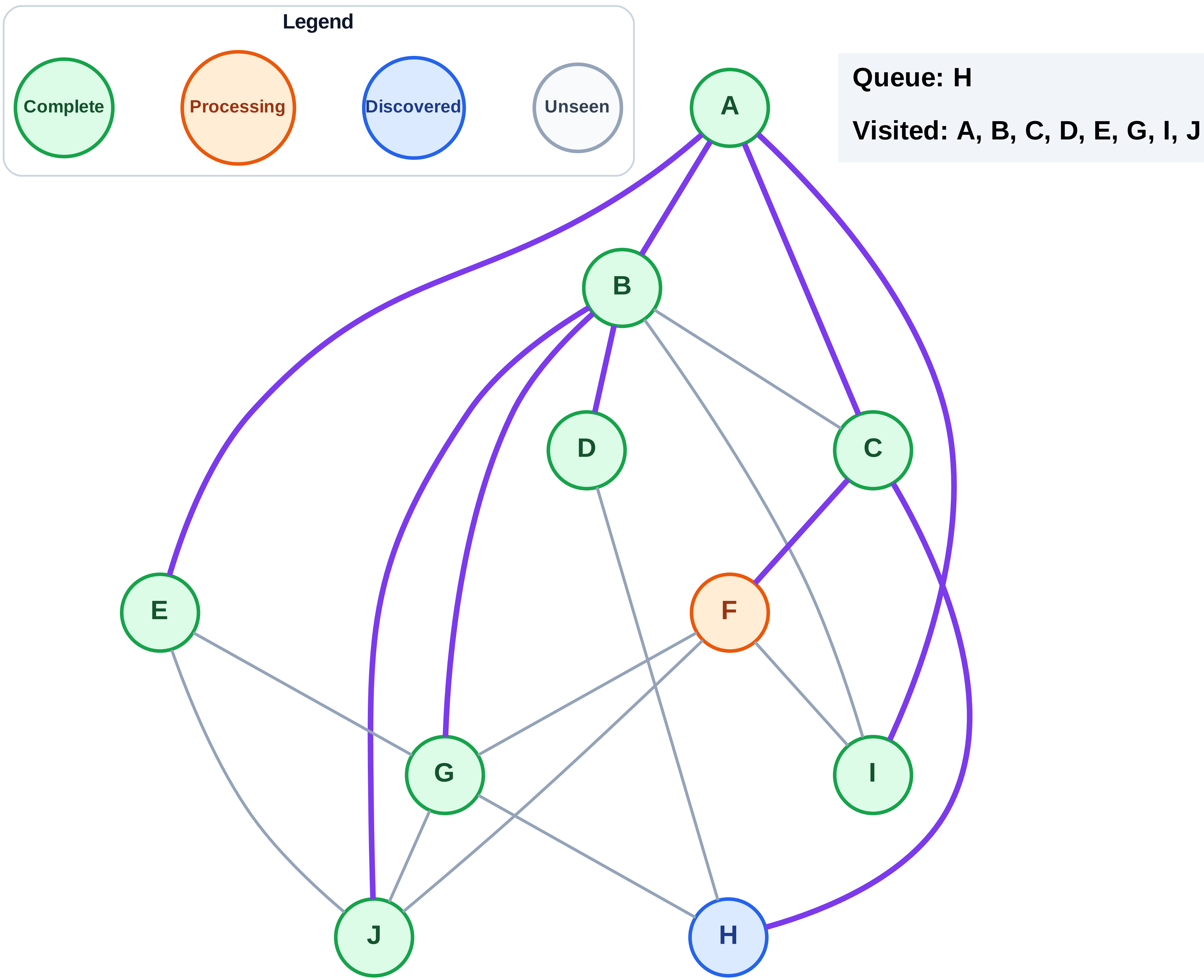
BFS Traversal

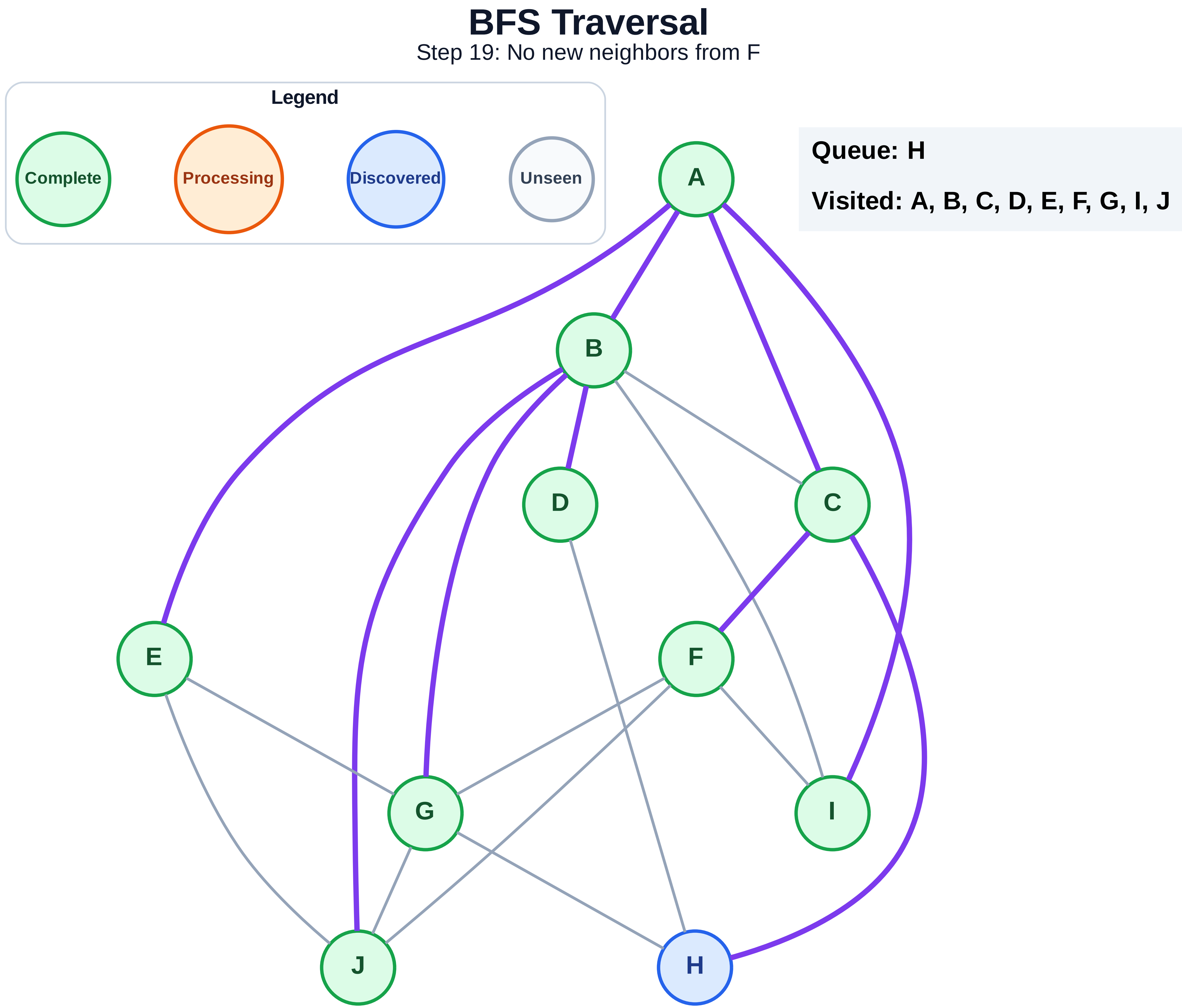
Step 17: No new neighbors from J



BFS Traversal

Step 18: Process node F

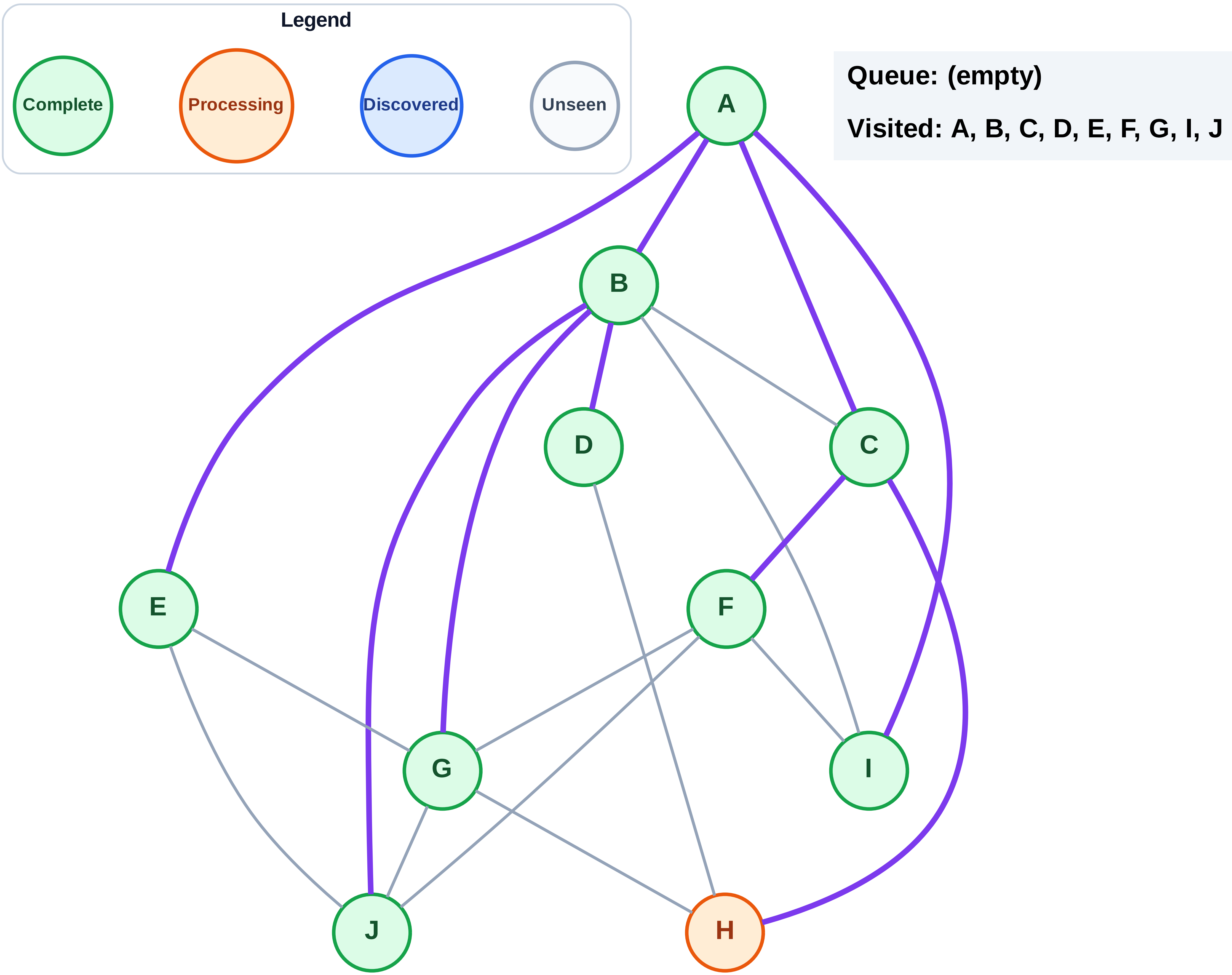


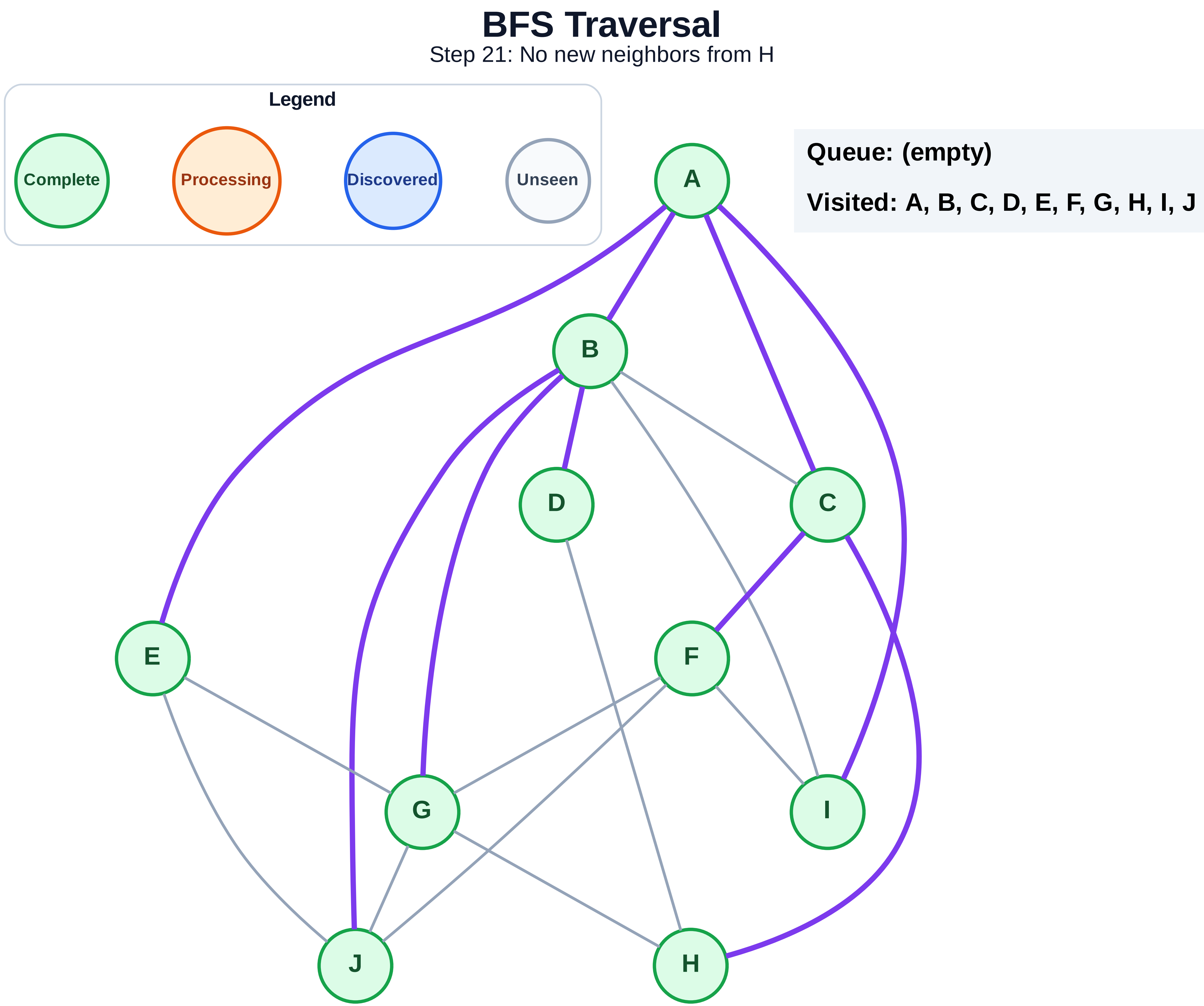


BFS Traversal
Step 19: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Queue: H
Visited: A, B, C, D, E, F, G, I, J
A
B
C
D
E
F
G
H
I
J

BFS Traversal

Step 20: Process node H

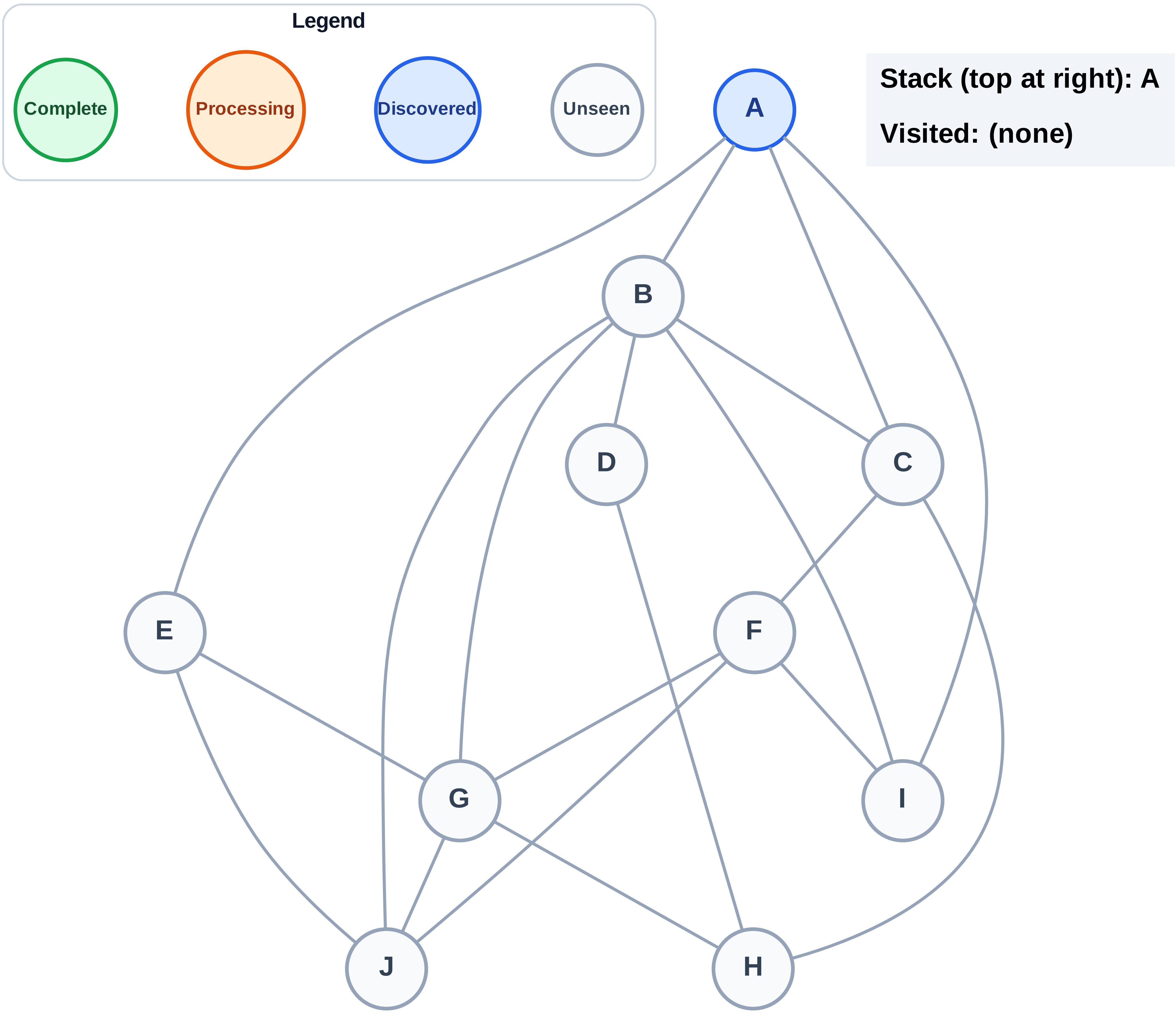




BFS Traversal
Step 21: No new neighbors from H
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
B
C
D
E
F
G
H
I
J

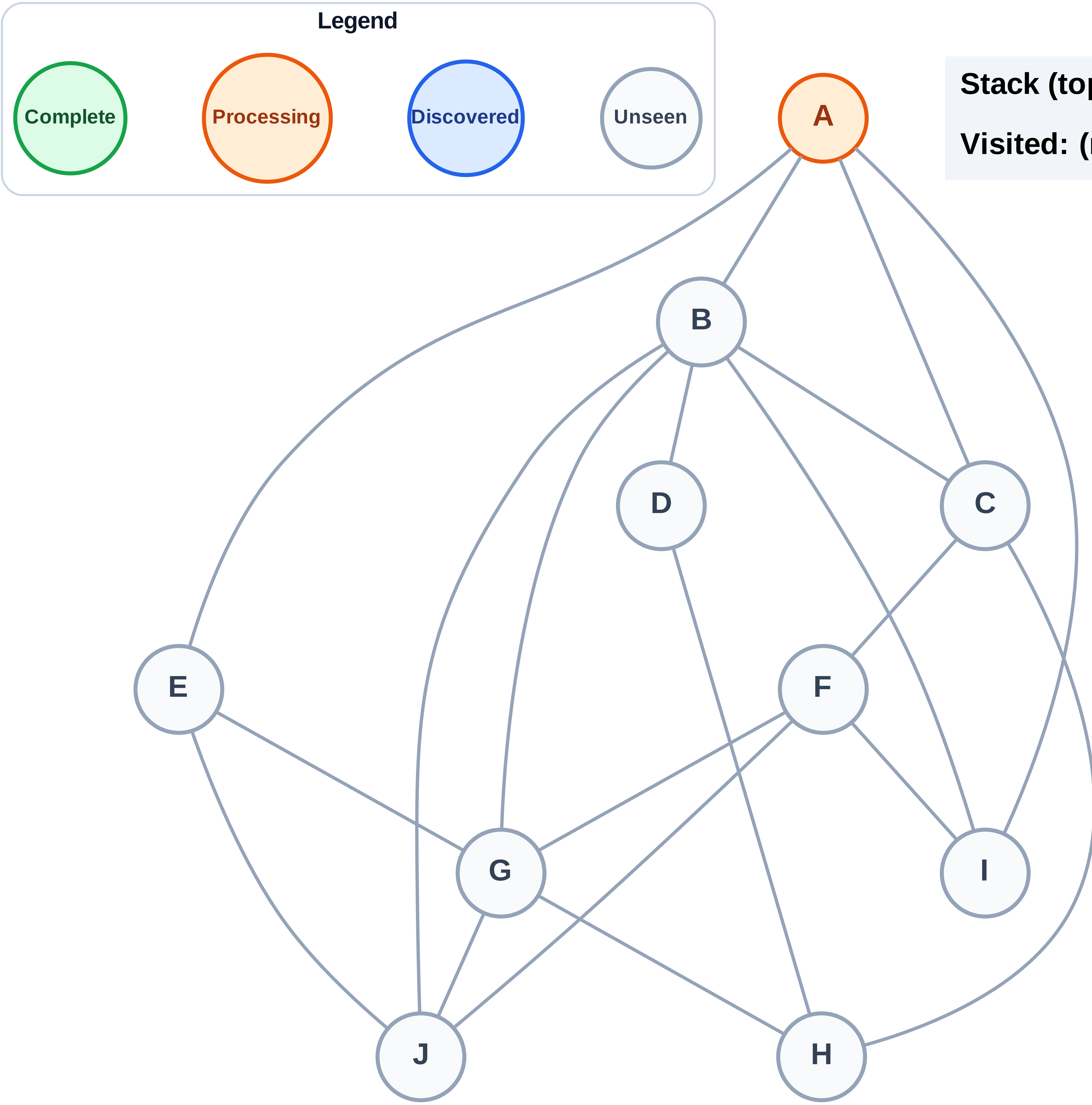
DFS Traversal

Step 1: Start at A



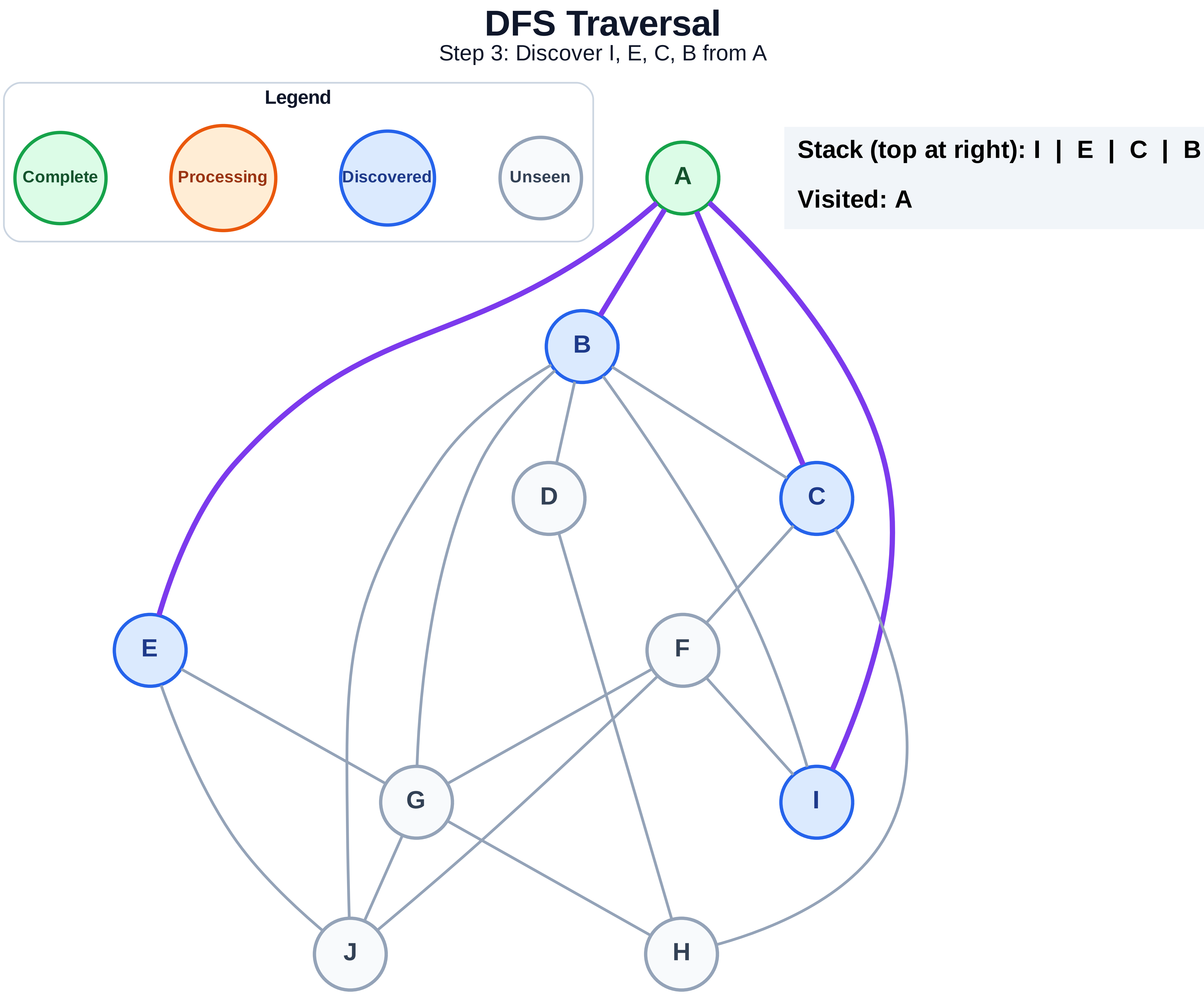
DFS Traversal

Step 2: Process node A



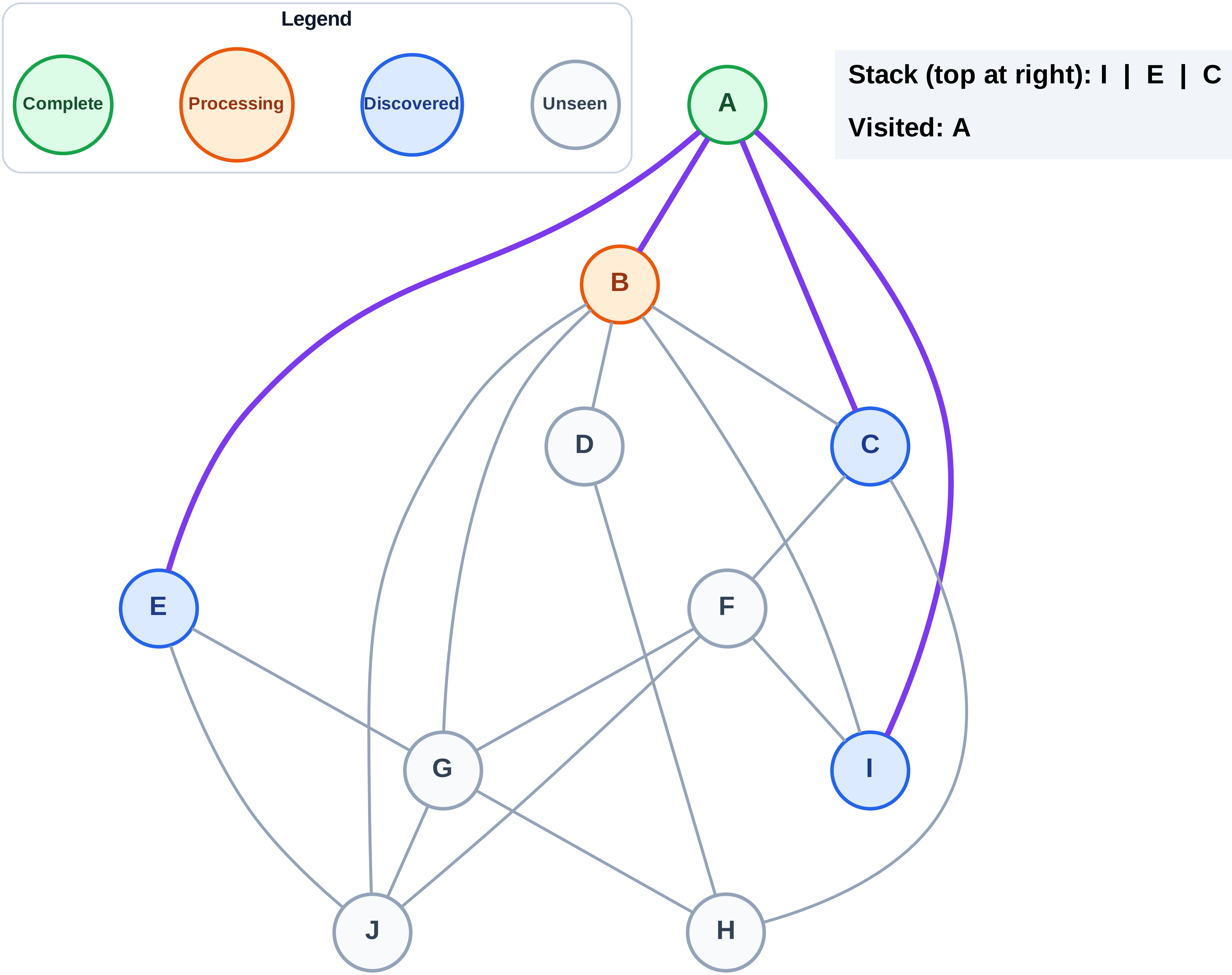
Stack (top at right): (empty)

Visited: (none)


DFS Traversal
Step 3: Discover I, E, C, B from A
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | E | C | B
Visited: A
A
B
C
D
E
F
G
H
I
J

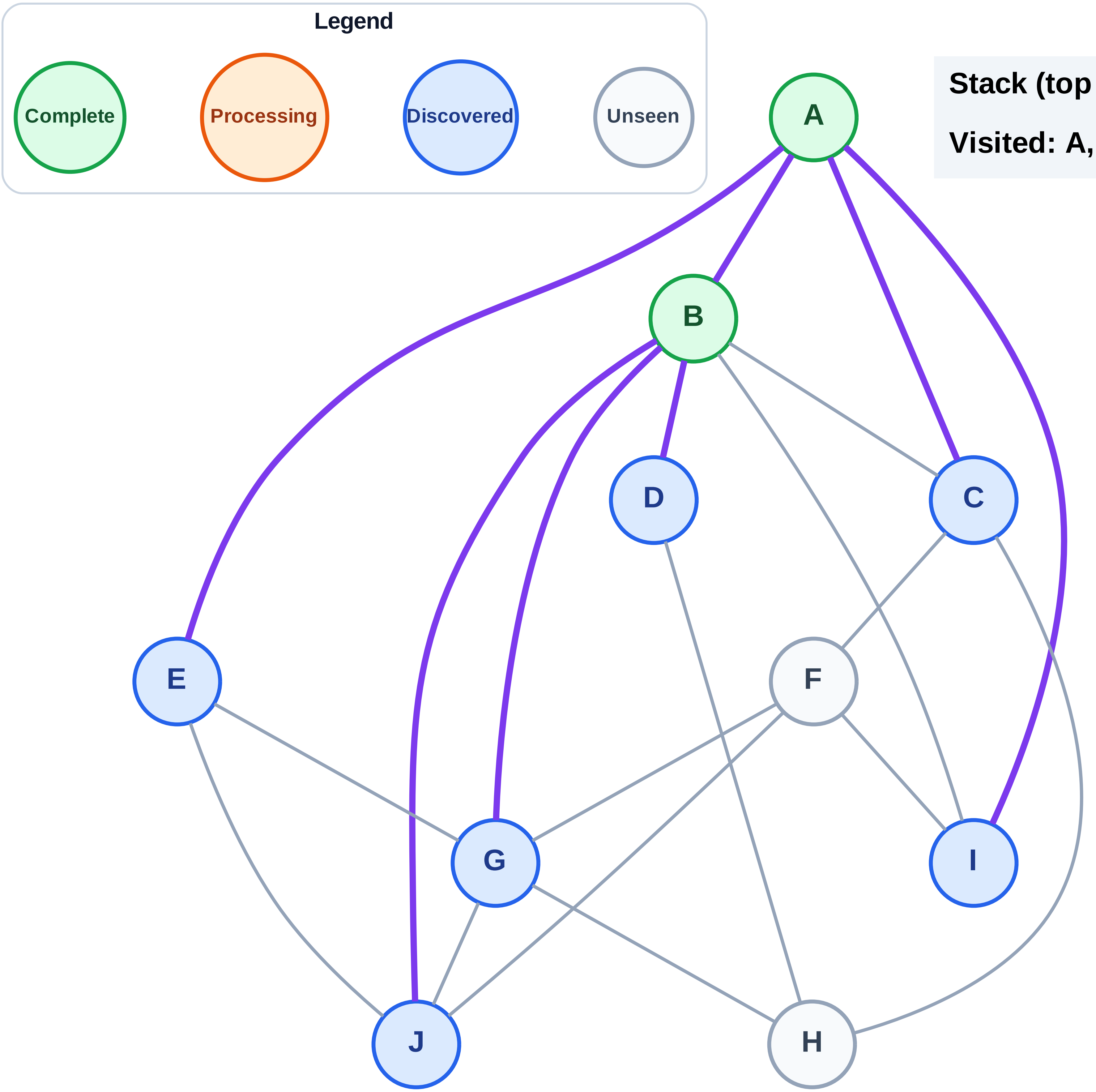
DFS Traversal

Step 4: Process node B



DFS Traversal

Step 5: Discover J, G, D from B

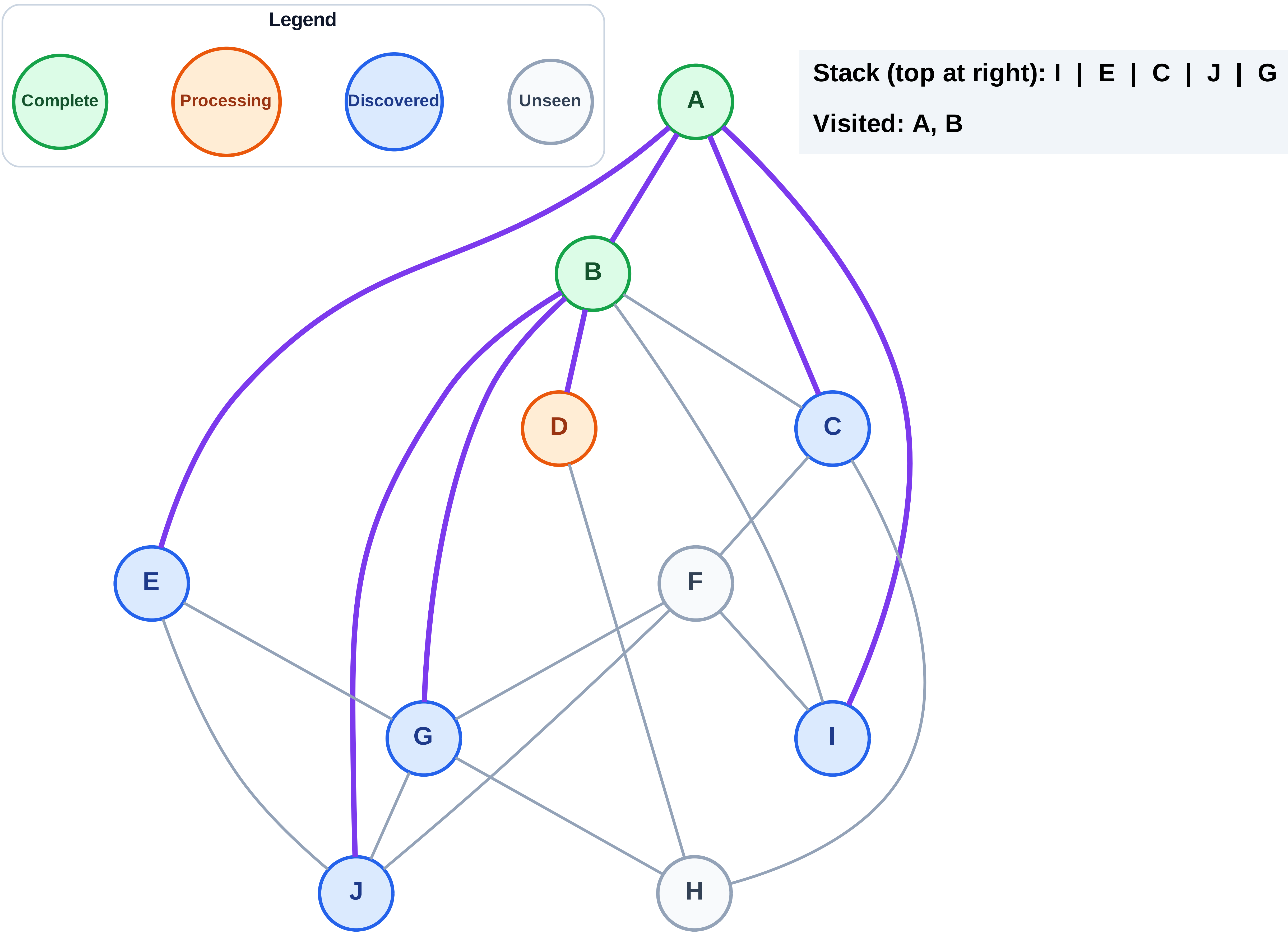


Stack (top at right): I | E | C | J | G | D

Visited: A, B

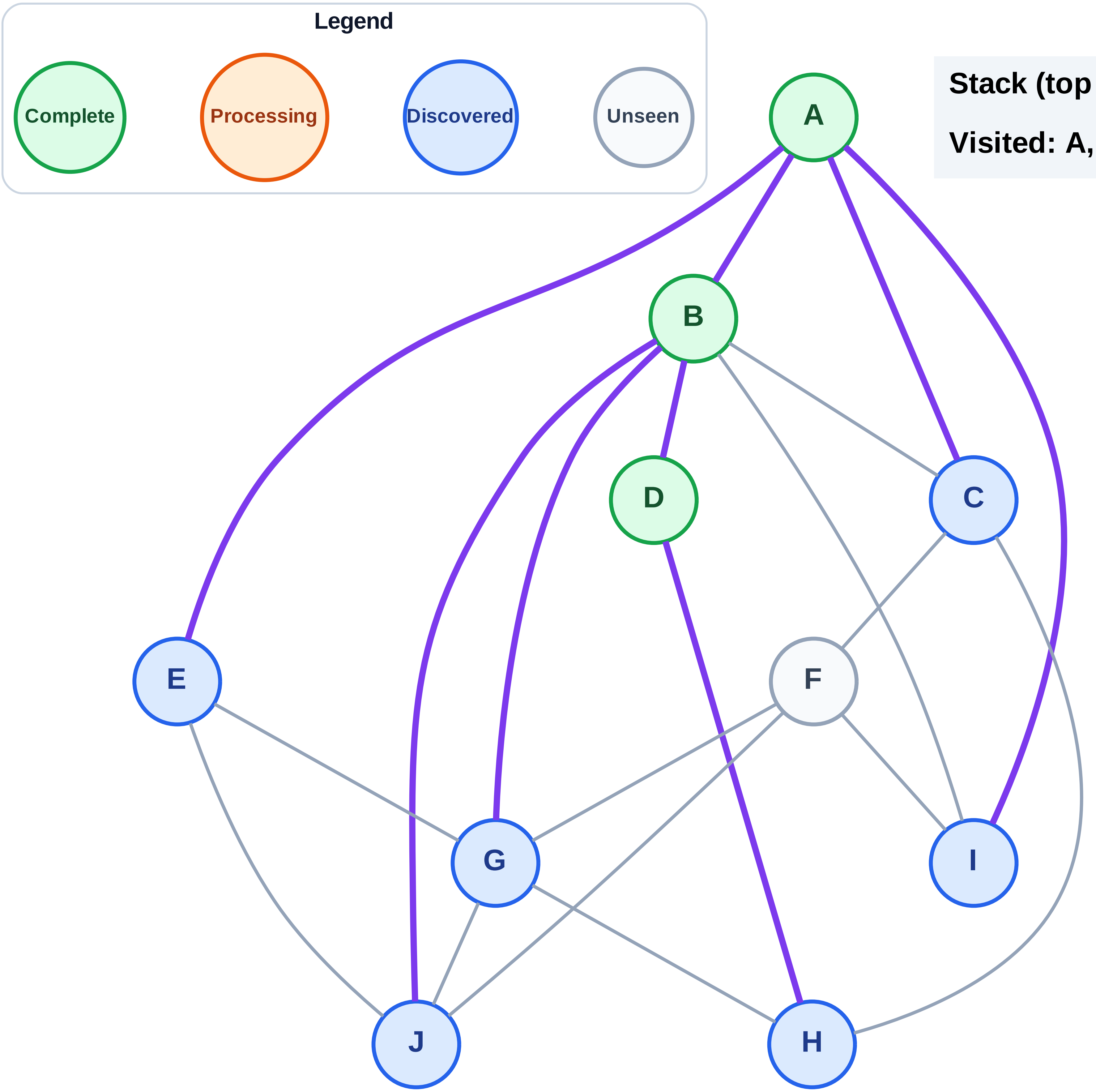
DFS Traversal

Step 6: Process node D



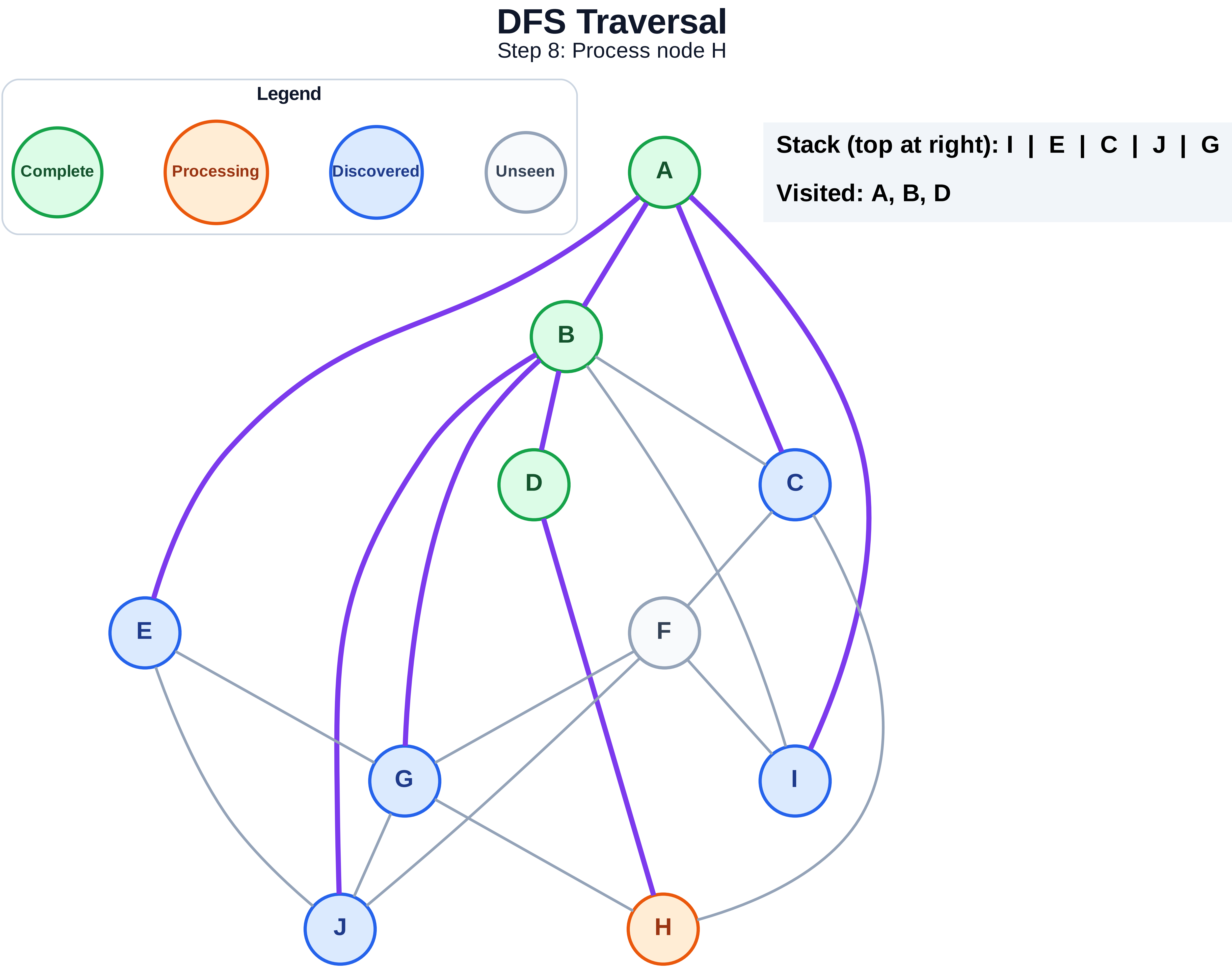
DFS Traversal

Step 7: Discover H from D



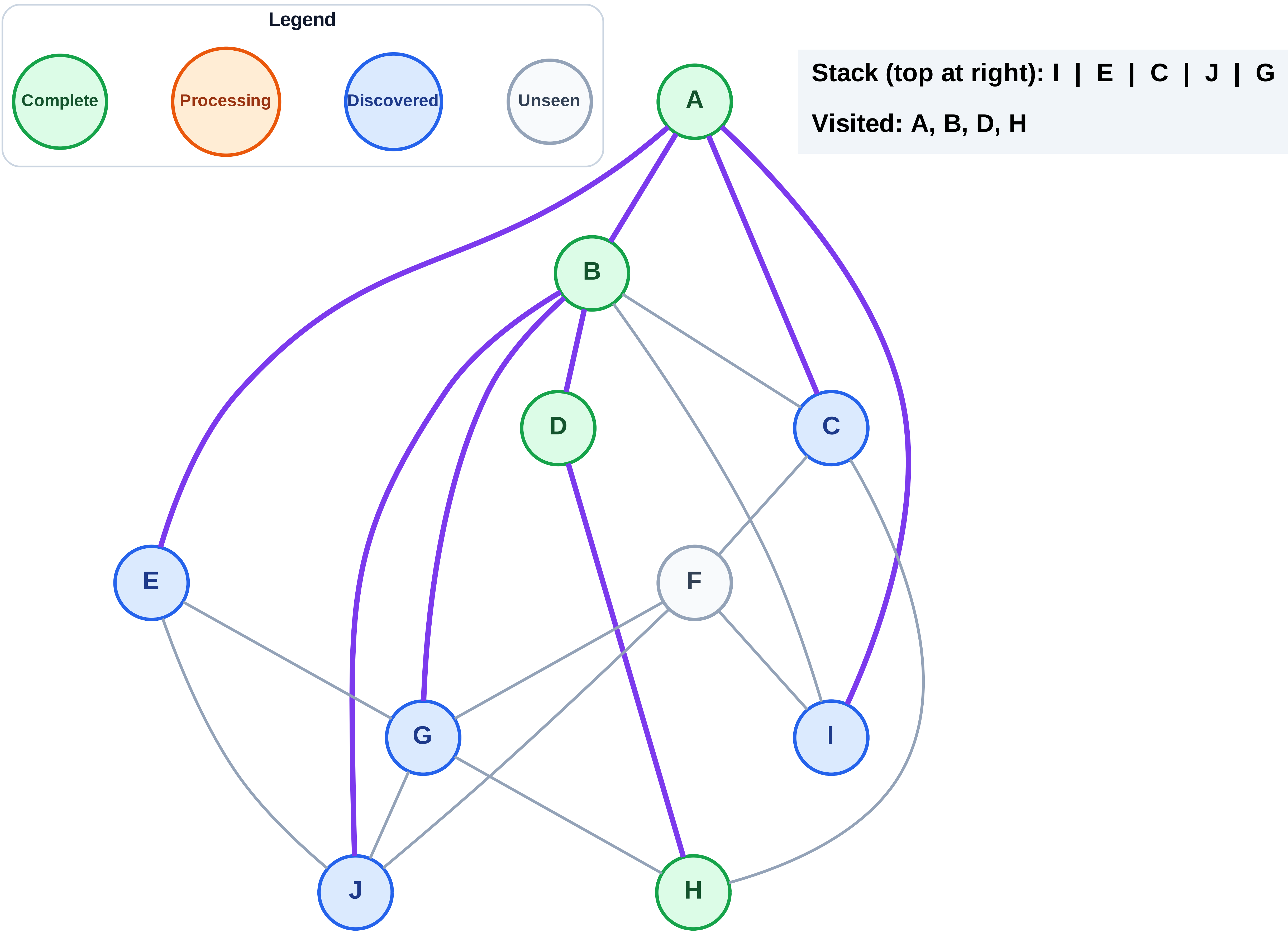
Stack (top at right): I | E | C | J | G | H

Visited: A, B, D


DFS Traversal
Step 8: Process node H
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | E | C | J | G
Visited: A, B, D
A
B
C
D
E
F
G
H
I
J

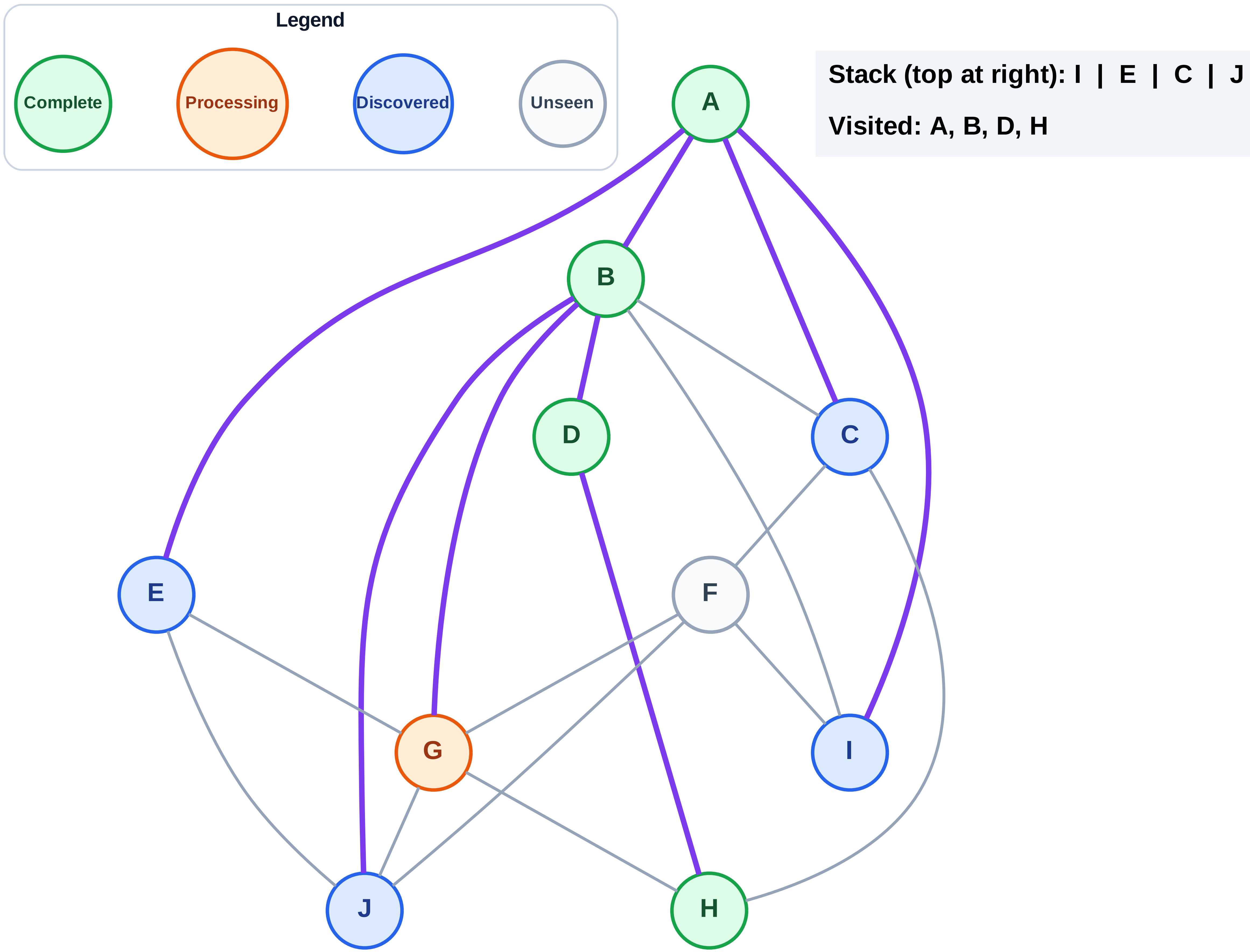
DFS Traversal

Step 9: No new neighbors from H



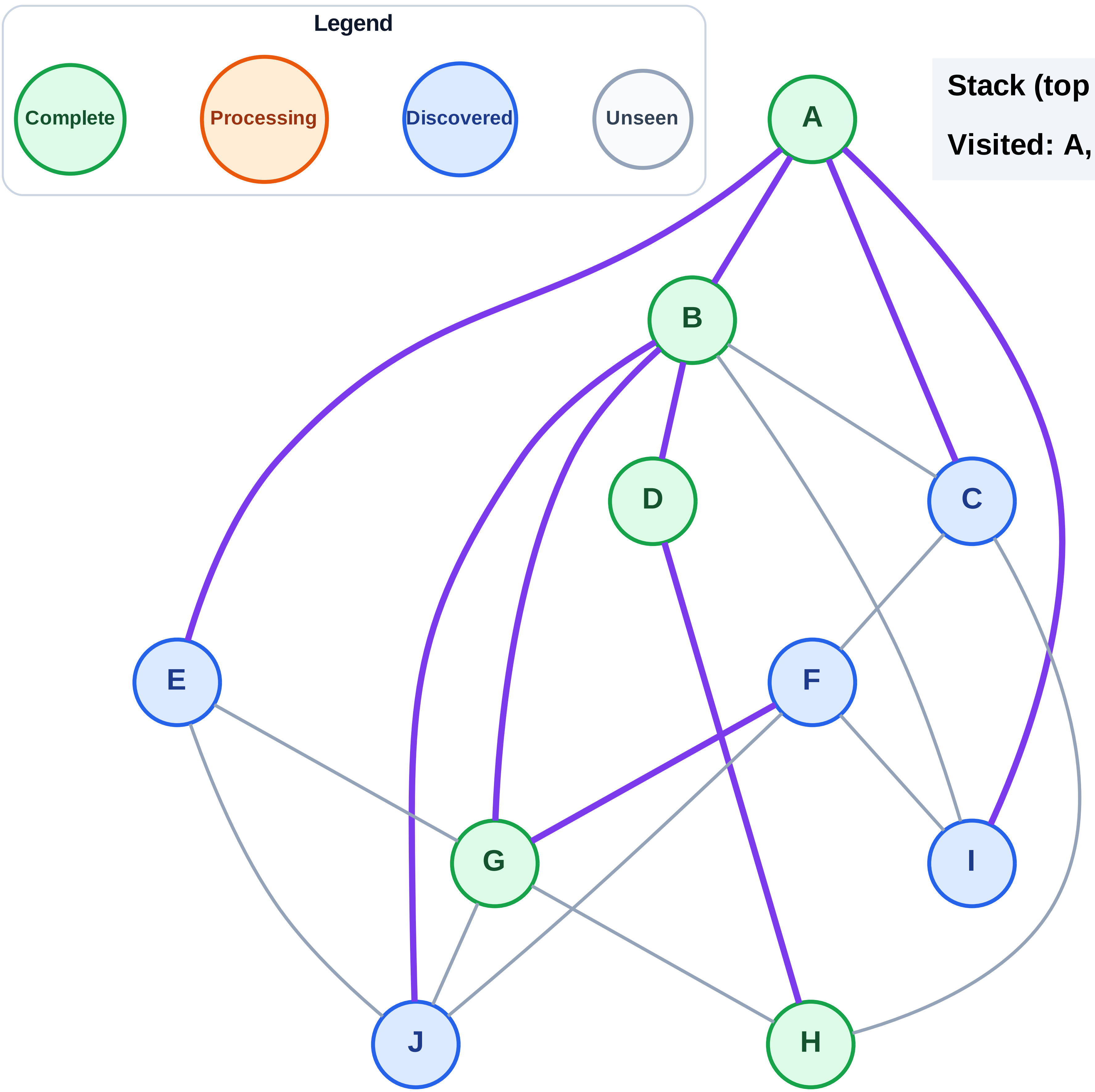
DFS Traversal

Step 10: Process node G



DFS Traversal

Step 11: Discover F from G

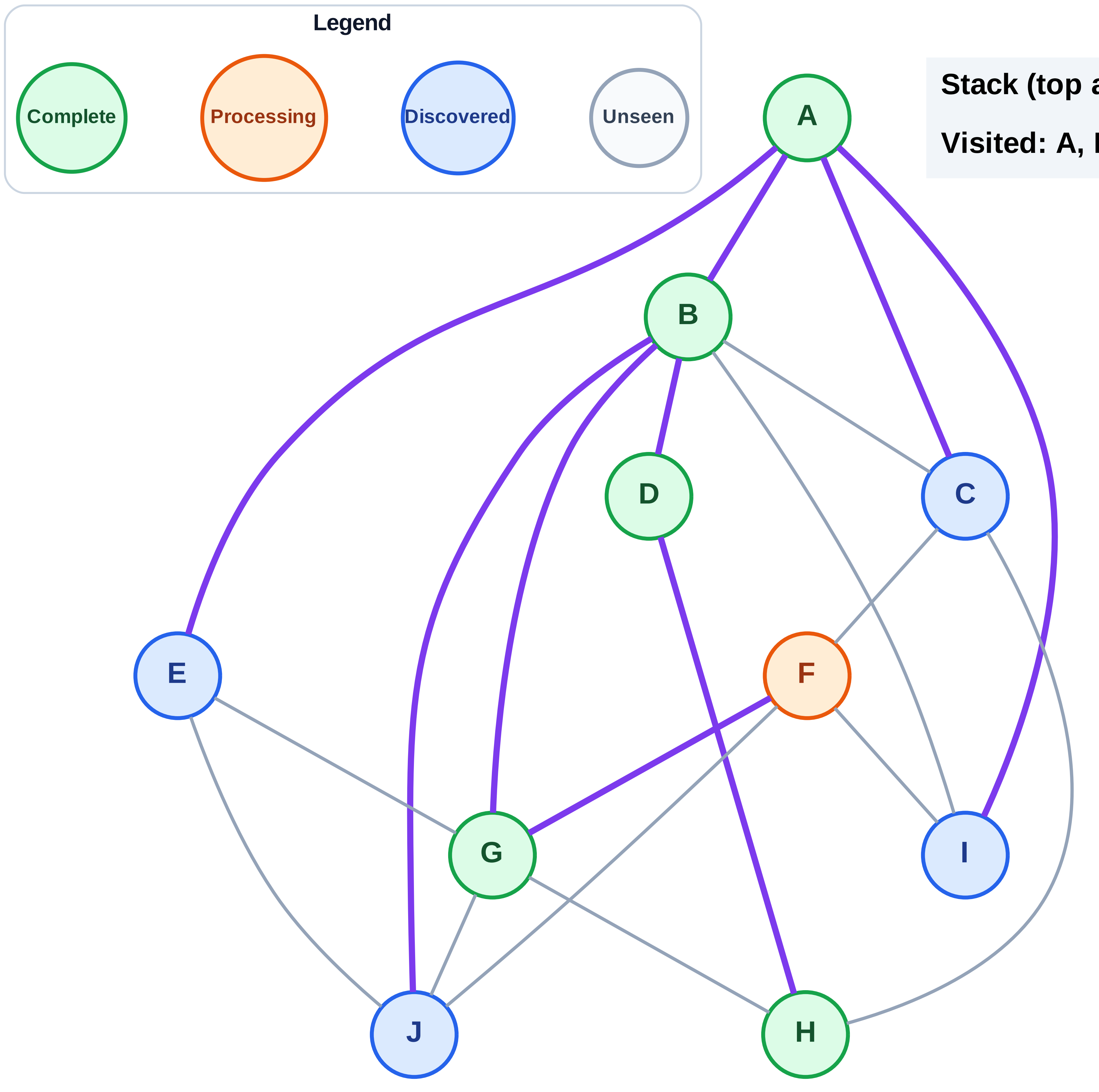


Stack (top at right): I | E | C | J | F

Visited: A, B, D, G, H

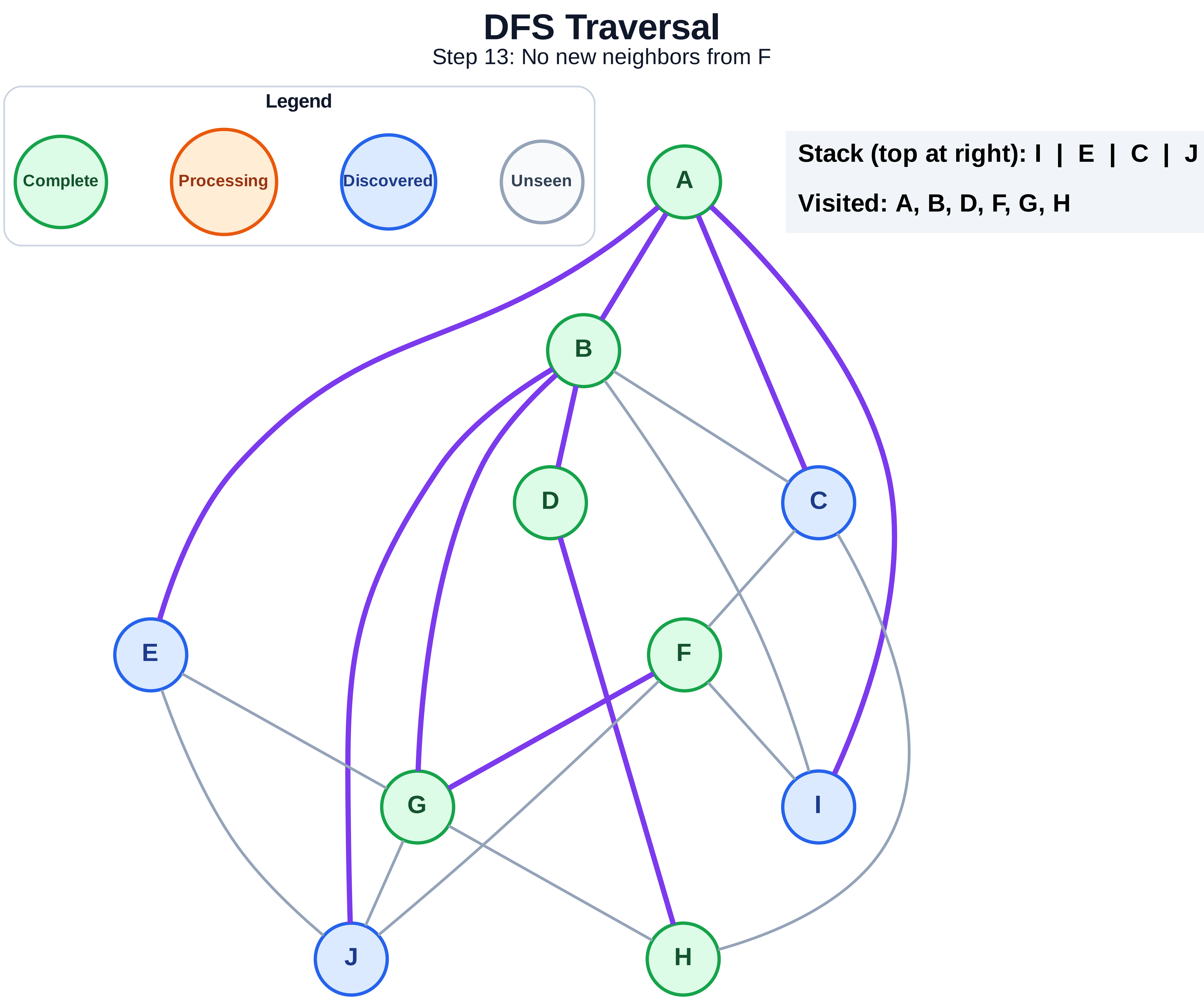
DFS Traversal

Step 12: Process node F



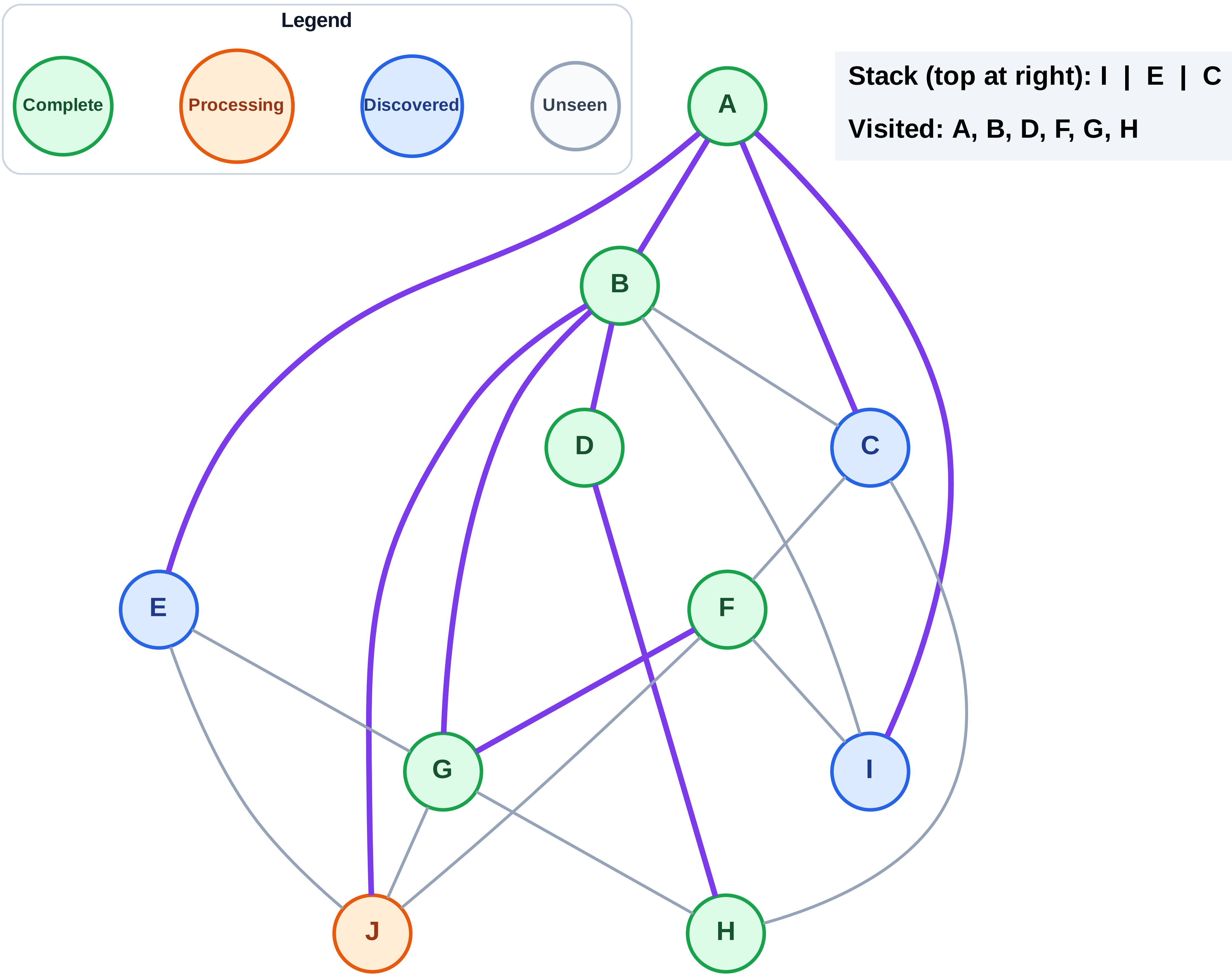
Stack (top at right): I | E | C | J

Visited: A, B, D, G, H


DFS Traversal
Step 13: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | E | C | J
Visited: A, B, D, F, G, H
A
B
C
D
E
F
G
H
I
J

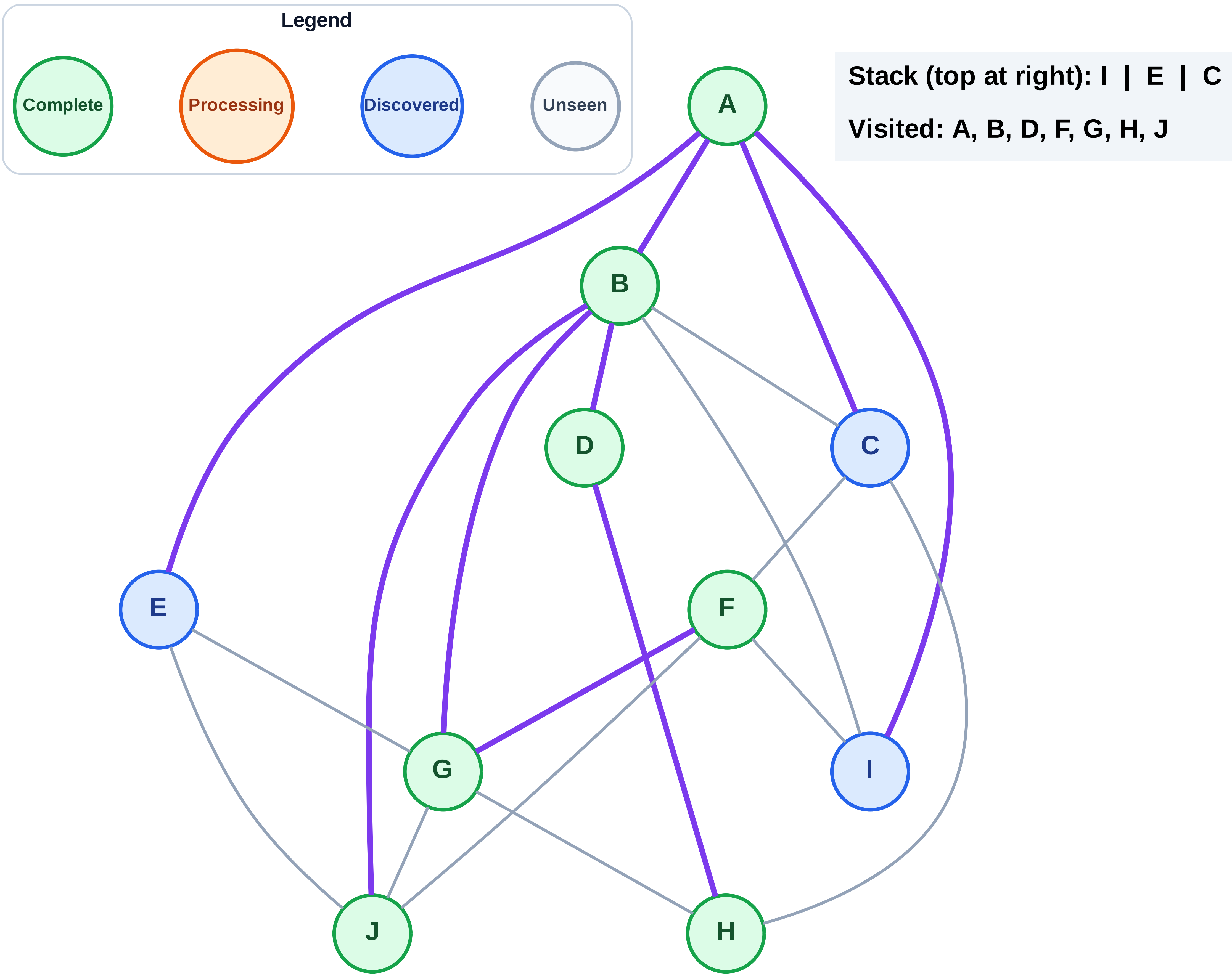
DFS Traversal

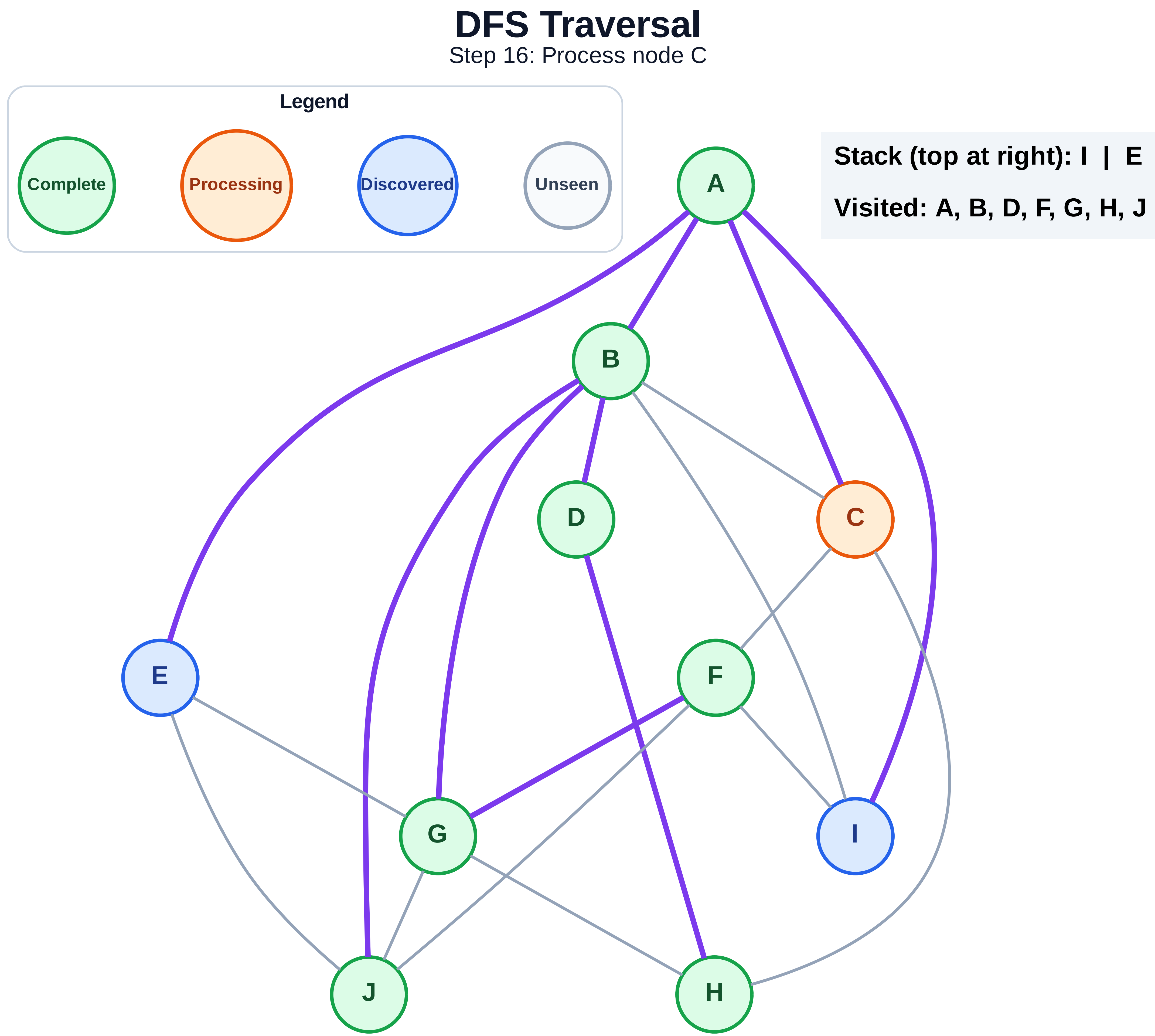
Step 14: Process node J



DFS Traversal

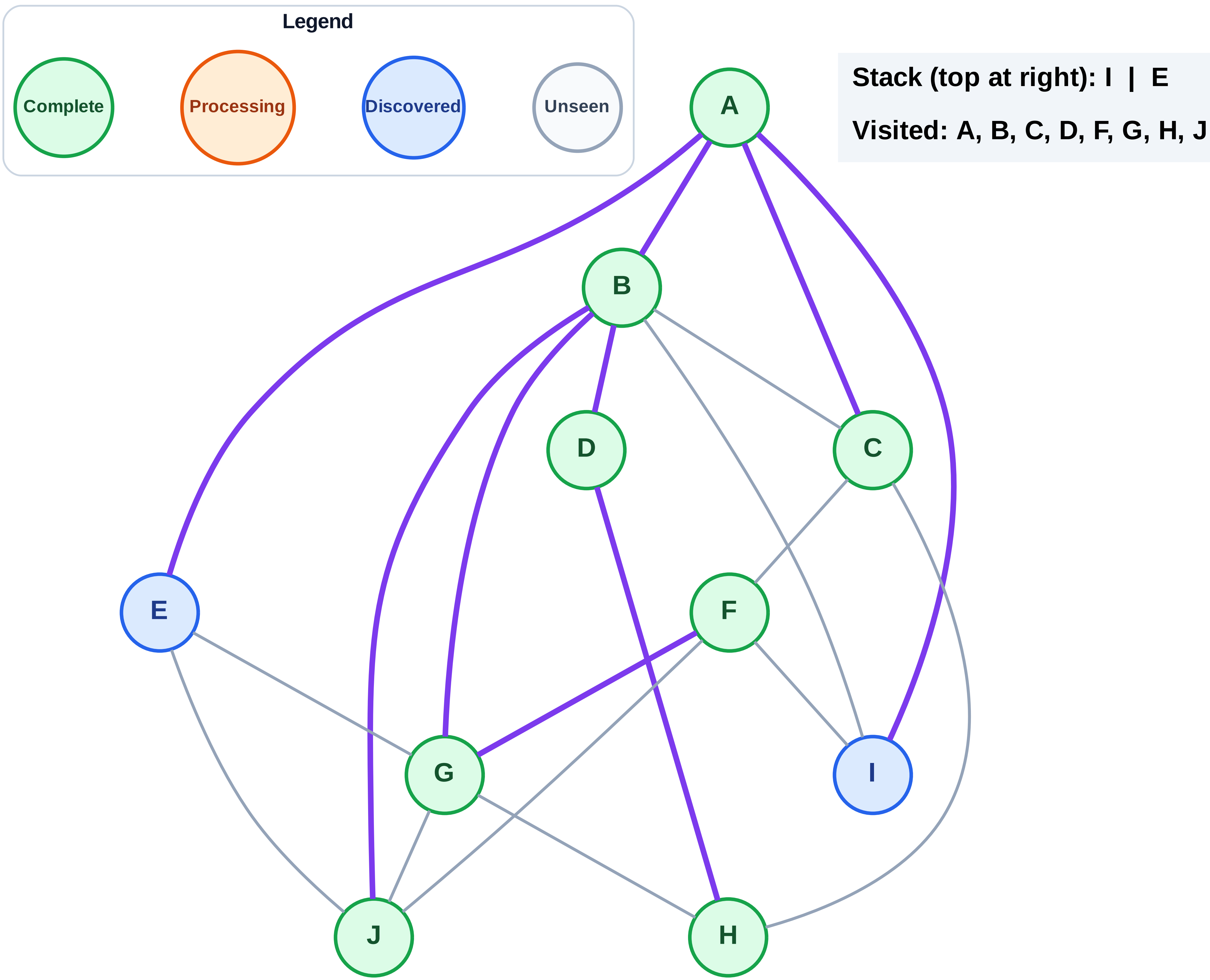
Step 15: No new neighbors from J

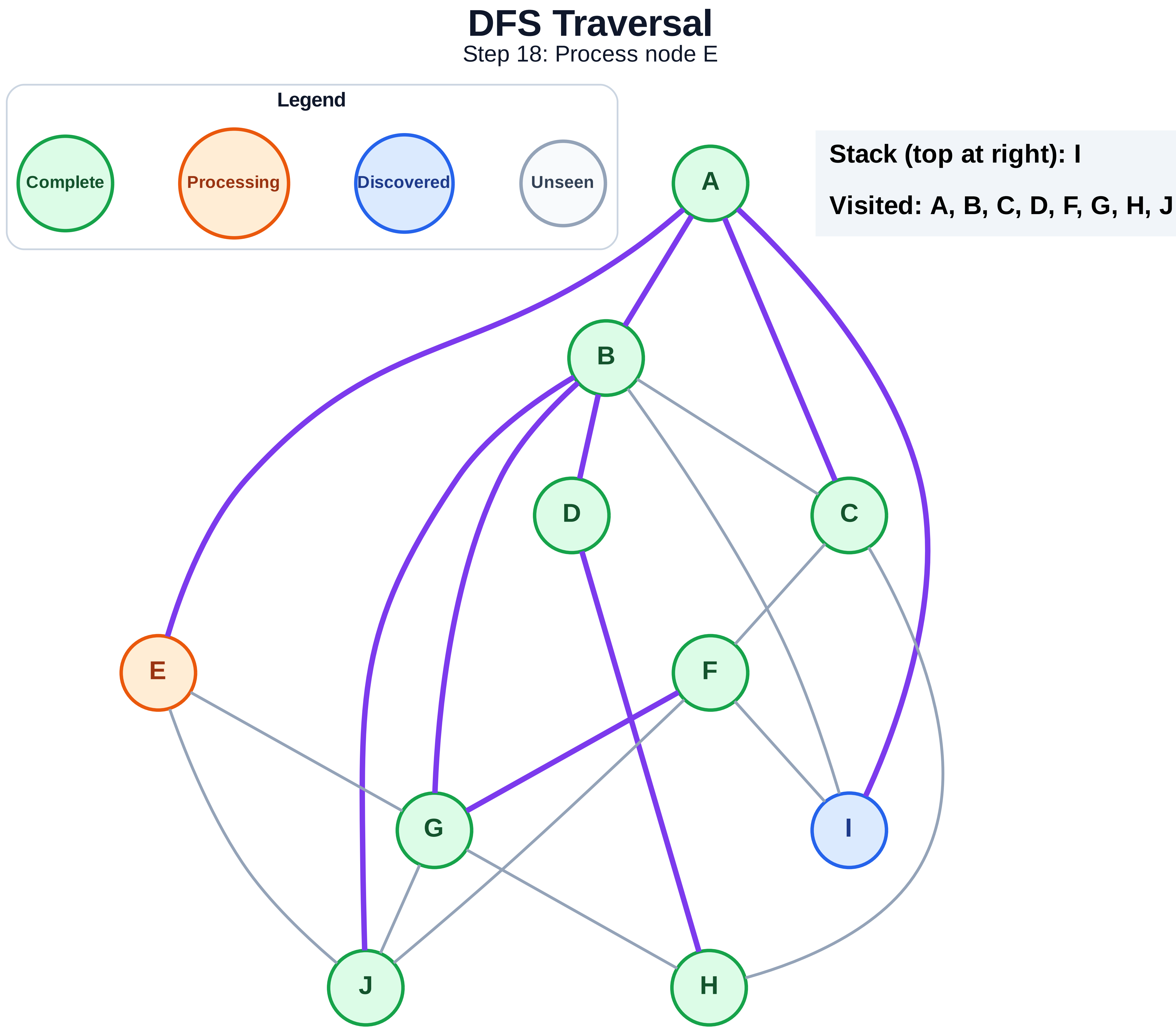



DFS Traversal
Step 16: Process node C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I | E
Visited: A, B, D, F, G, H, J
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 17: No new neighbors from C

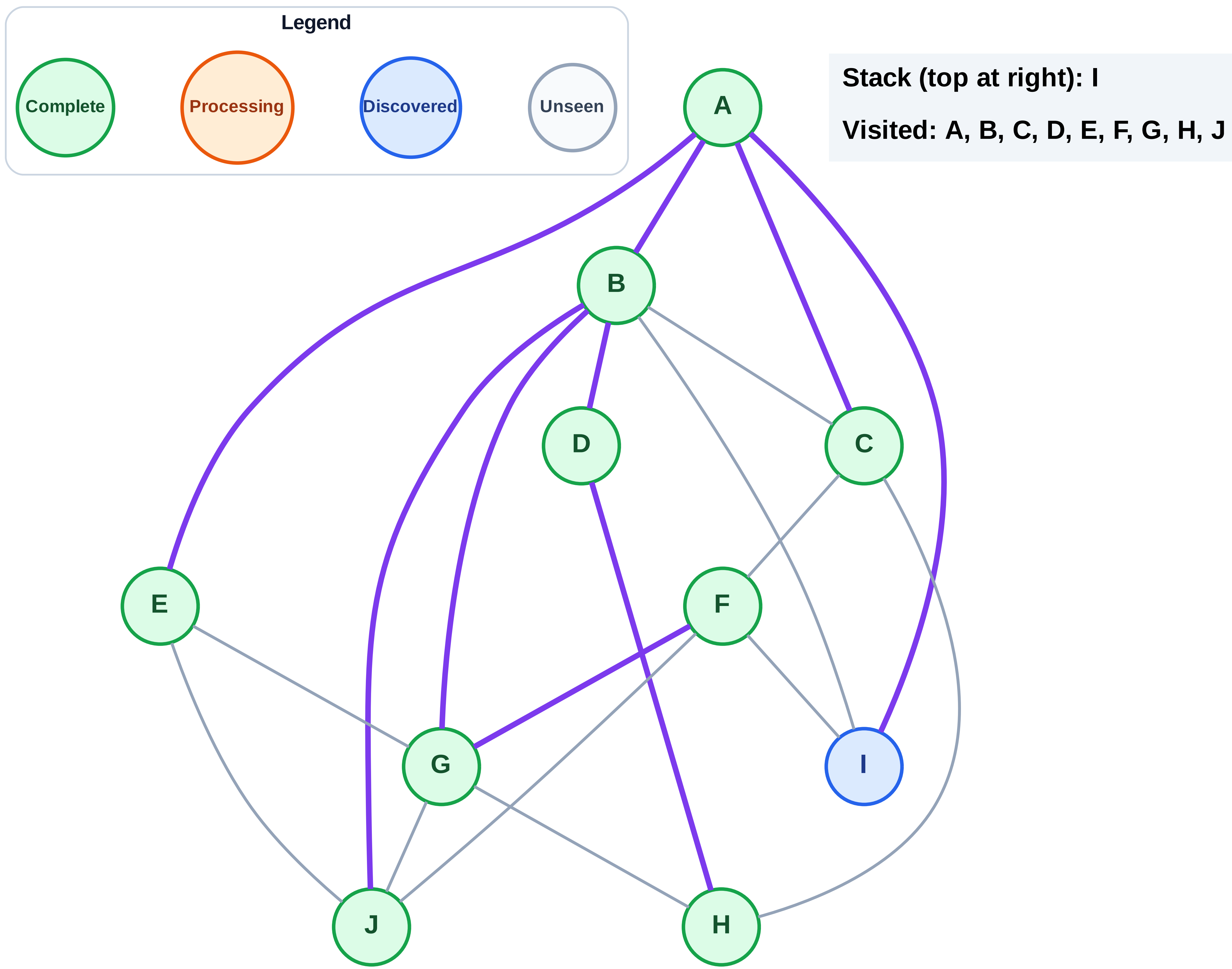




DFS Traversal
Step 18: Process node E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I
Visited: A, B, C, D, F, G, H, J
A
B
C
D
E
F
G
H
I
J

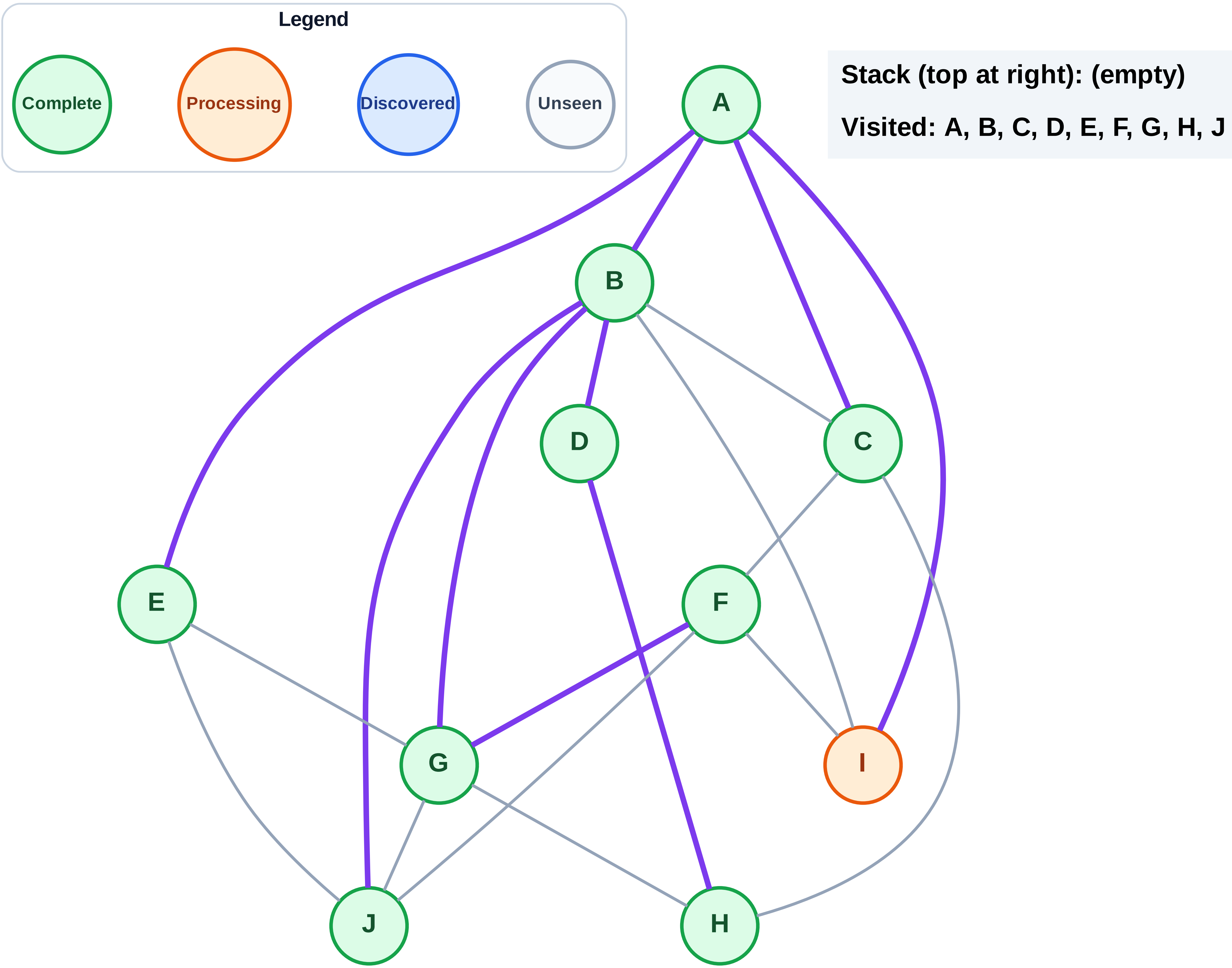
DFS Traversal

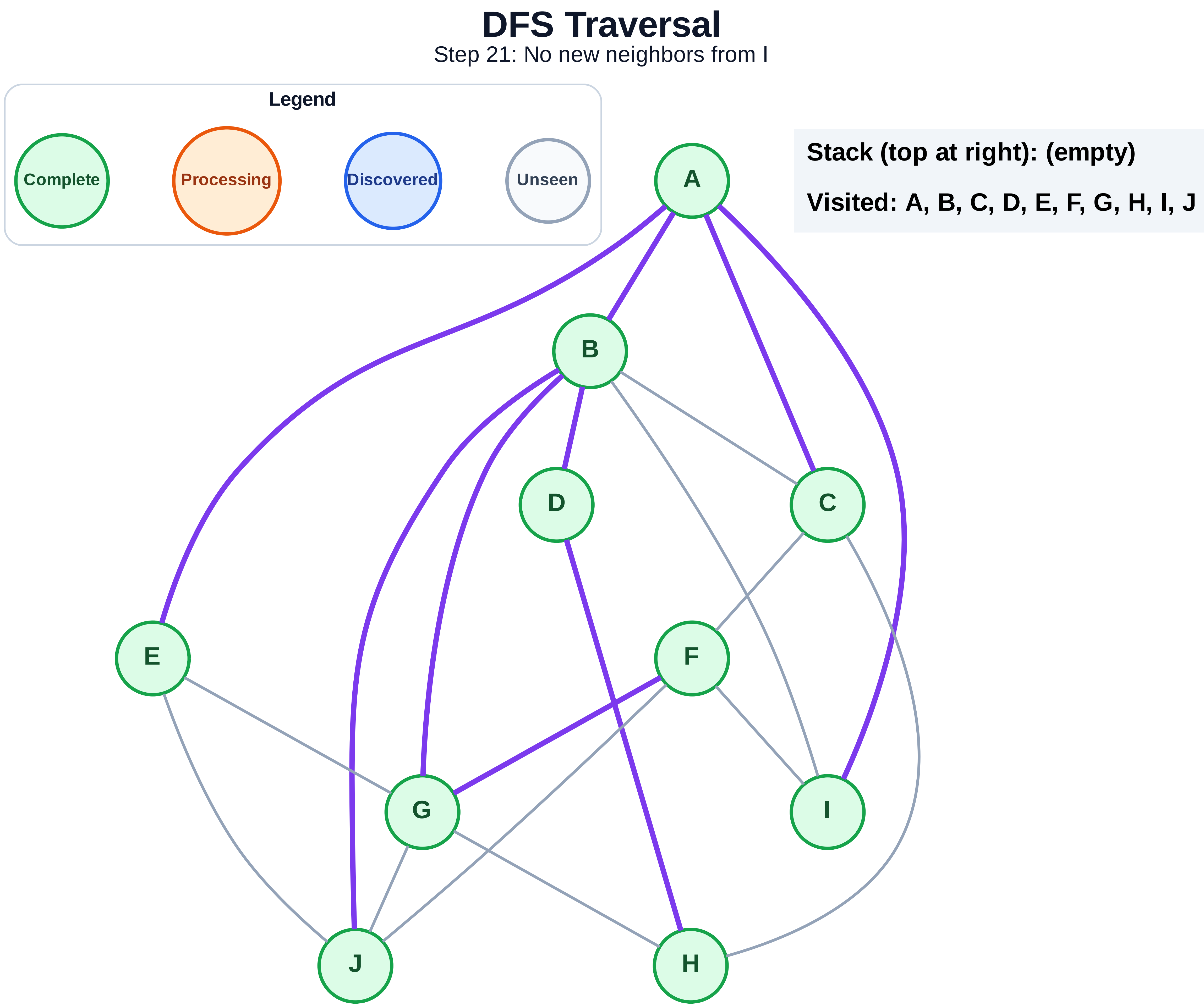
Step 19: No new neighbors from E



DFS Traversal

Step 20: Process node I




DFS Traversal
Step 21: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
B
C
D
E
F
G
H
I
J